



NTNU – Trondheim
Norwegian University of
Science and Technology

TET 4565 -Electricity Markets and Energy System Planning, Specialisation Course

**Decomposition techniques, Bender's
decomposition (theory and application)**

Special thanks to professor Jalal Kazempour at DTU

Hossein Farahmand

Professor

Autumn 2026

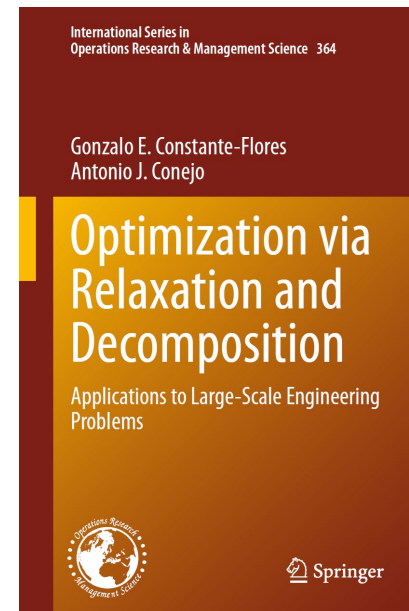
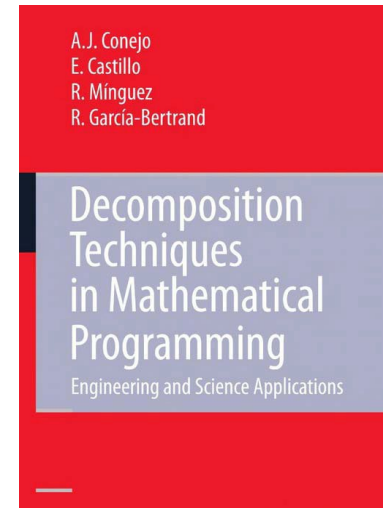
Overview

- Decomposition techniques
- Explain the functioning of Benders' decomposition (concept)
- Implement it to an illustrative problem
- **Implementation of Bender's decomposition to two-stage stochastic programming**

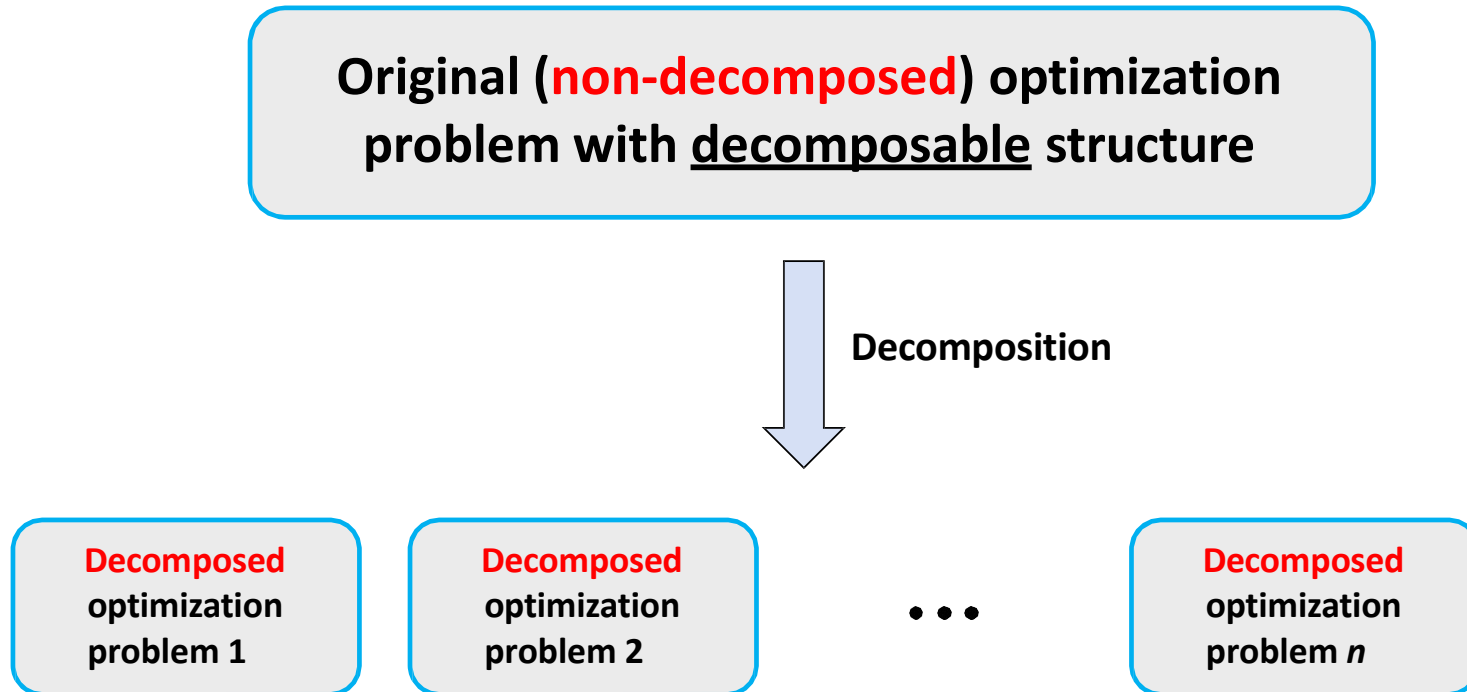
Reference

A. J. Conejo, E. Castillo, R. Minguez, and R. Garcia-Bertrand, *Decomposition Techniques in Mathematical Programming: Engineering and Science Applications*. Berlin, Germany: Springer, 2006.
(Chapter 3, 3.3)

Chapter 5
Solving Optimization Problems with
Complicating Variables



Decomposition: main idea



Each decomposed problem is **easier to solve** than the original (non-decomposed) problem!

Decomposable structure

Optimization problems with decomposable structure:

- Problems with complicating variable(s):

The original problem is decomposed if **complicating variables** are **fixed** to given values!

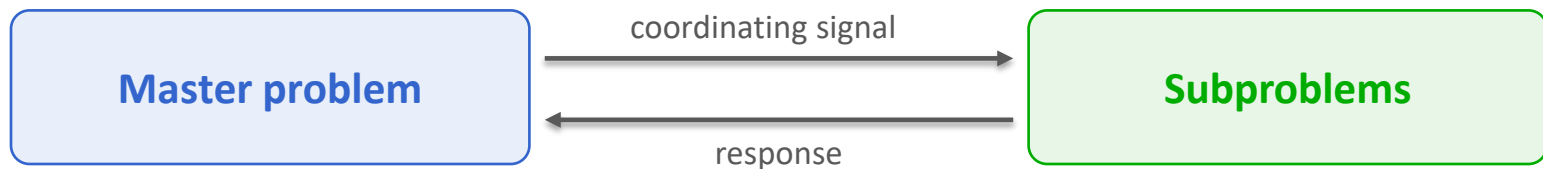
- Problems with complicating constraint(s):

The original problem is decomposed if **complicating constraints** are **relaxed** (removed)!

In the literature, “complicating” variables/constraints are also called as “coupling” or “linking” variables/constraint!

Features of decomposition techniques

- Iterative solution techniques
- Original optimization problem decomposes to:
 - A single **master** problem (not necessarily an optimization problem)
 - A set of **subproblems**



Iterate: the two levels exchange information until they agree on the solution.

Illustrative example

Single-node single-year (static) generation expansion problem, considering two existing generators (units 1 and 2) and one candidate generator (unit 3)

Linking variable or complicating variable

$$\text{Minimise } \underbrace{c_3^{inv} x_3}_{\text{Expansion cost of candidate unit}} + \underbrace{\sum_h [c_1 g_{h,1} + c_2 g_{h,2} + c_3 g_{h,3}]}_{\text{Operational cost of (existing and candidate) units}}$$

subject to

$$0 \leq g_{h,1} \leq \bar{g}_1 \quad \forall h$$

$$0 \leq g_{h,2} \leq \bar{g}_2 \quad \forall h$$

$$0 \leq g_{h,3} \leq x_3 \quad \forall h$$

$$g_{h,1} + g_{h,2} + g_{h,3} = D_{h,1} \quad \forall h$$

$$x_3 \geq 0$$

Illustrative example- decomposed problem

We fixed $x_3 \rightarrow x_3^{fixed}$ and remove the fixed term from the objective function.
The subproblems will be:

$$\left\{ \begin{array}{l} \text{Minimise } c_1 g_{h,1} + c_2 g_{h,2} + c_3 g_{h,3} \\ \quad g_{h,1}, g_{h,2}, g_{h,3} \\ \text{subject to,} \\ 0 \leq g_{h,1} \leq \bar{g}_1 \\ 0 \leq g_{h,2} \leq \bar{g}_2 \\ 0 \leq g_{h,3} \leq x_3^{fixed} \\ g_{h,1} + g_{h,2} + g_{h,3} = D_{h,1} \end{array} \right\} \forall h$$

The idea behind Benders' decomposition

Questions for this part

- How can we find the optimum when the cost profile is not available?
- What is a cut, and what do we need to draw one?
- Why do more cuts take us to the optimal point?

Seminal paper of Benders

Numerische Mathematik 4, 238—252 (1962)

Partitioning procedures for solving mixed-variables programming problems*

By

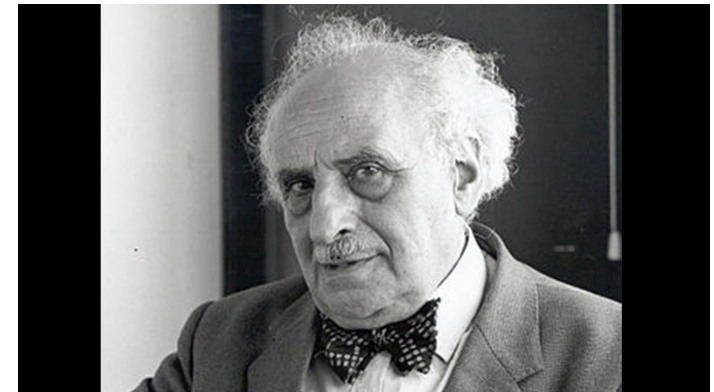
J. F. BENDERS**

I. Introduction

In this paper two slightly different procedures are presented for solving mixed-variables programming problems of the type

$$\max \{c^T x + f(y) \mid Ax + F(y) \leq b, x \in R_p, y \in S\}, \quad (1.1)$$

where $x \in R_p$ (the p -dimensional Euclidean space), $y \in R_q$, and S is an arbitrary subset of R_q . Furthermore, A is an (m, p) matrix, $f(y)$ is a scalar function and



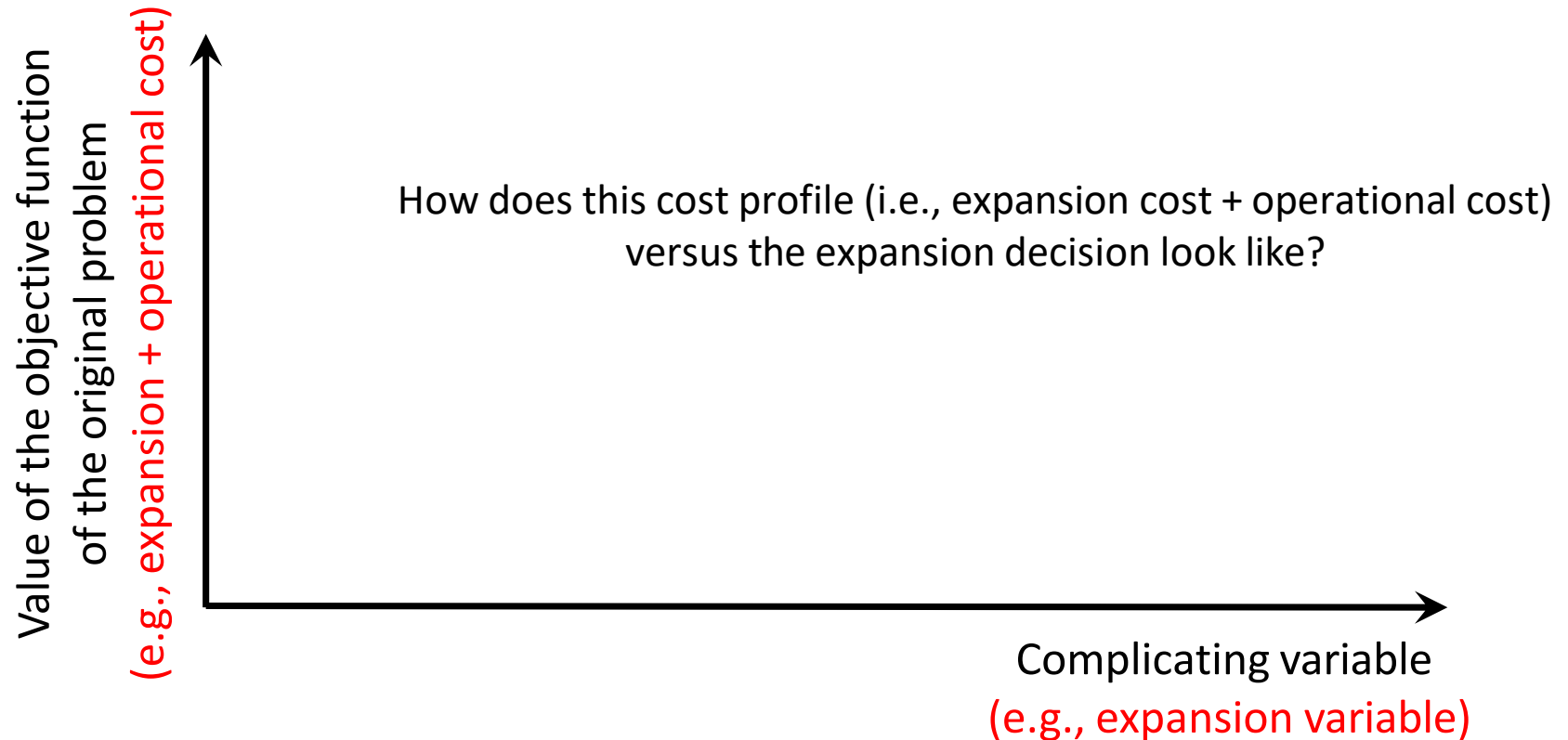
Jacobus Franciscus (Jacques) Benders

(1 June 1924 - 9 January 2017)

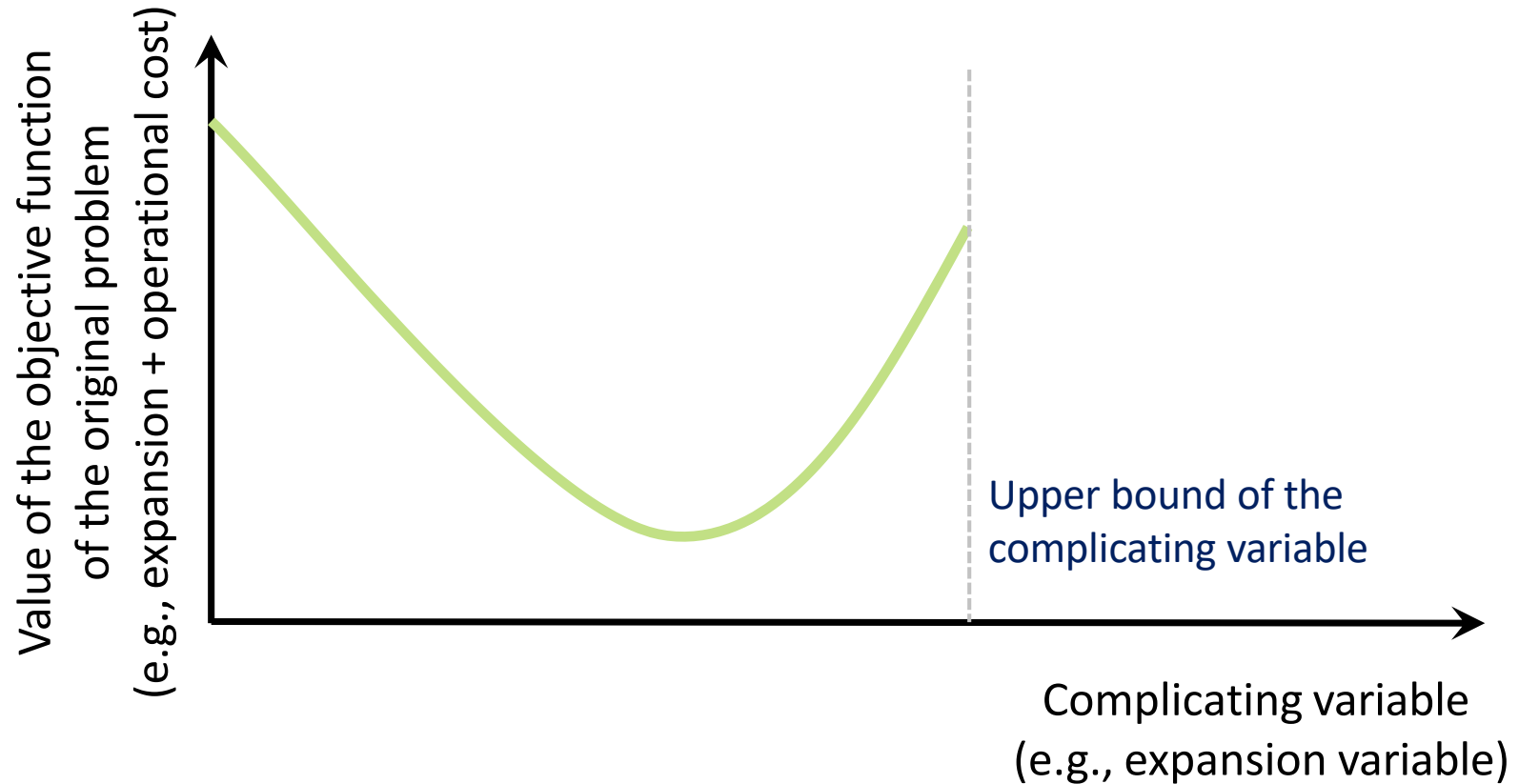
https://en.wikipedia.org/wiki/Jacques_F._Benders

J. F. Benders, "Partitioning procedures for solving mixed-variables programming problems," *Numerische Mathematik*, vol. 4, no. 1, pp. 238–252, 1962.

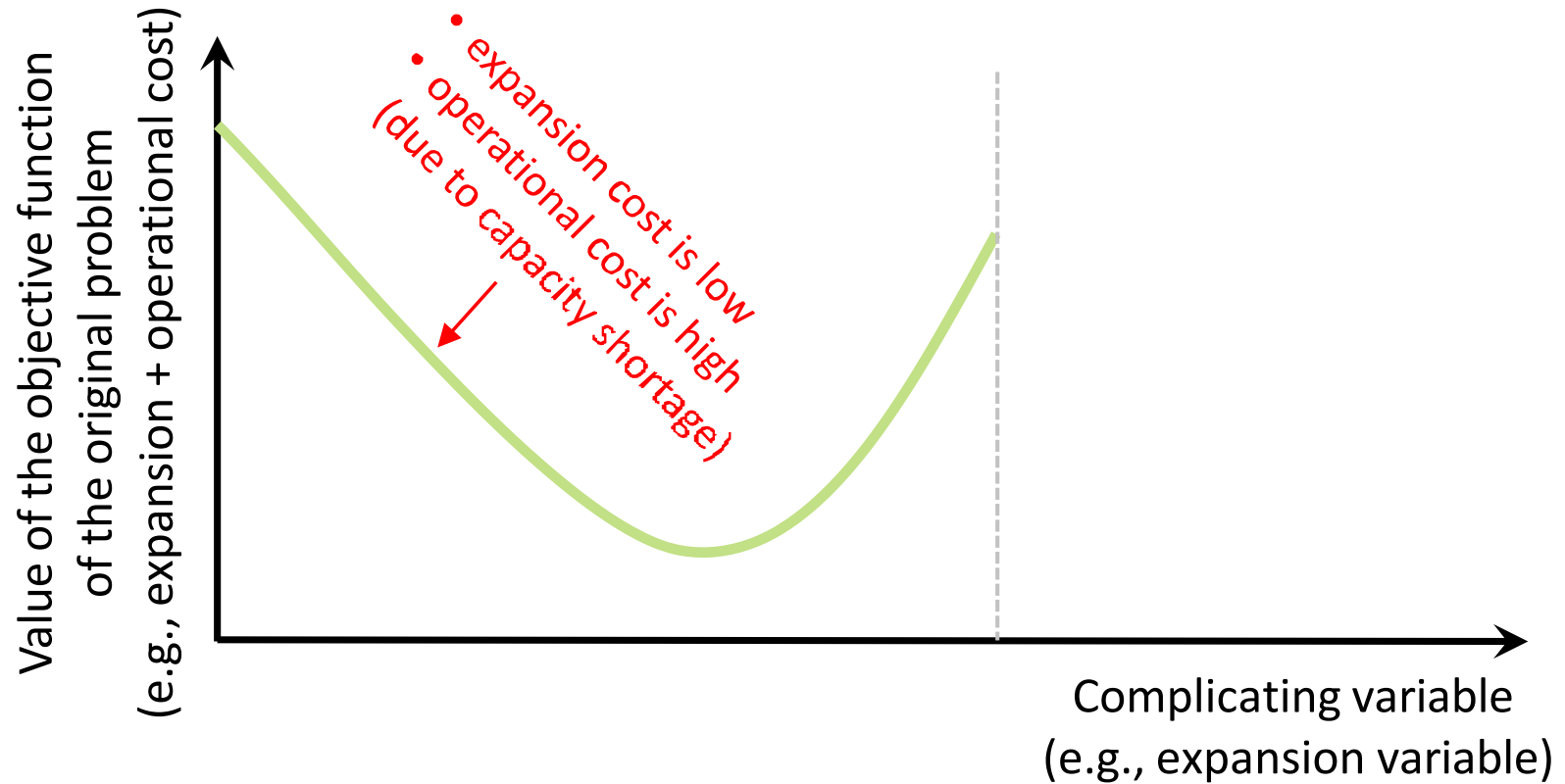
Concept-capacity expansion problem



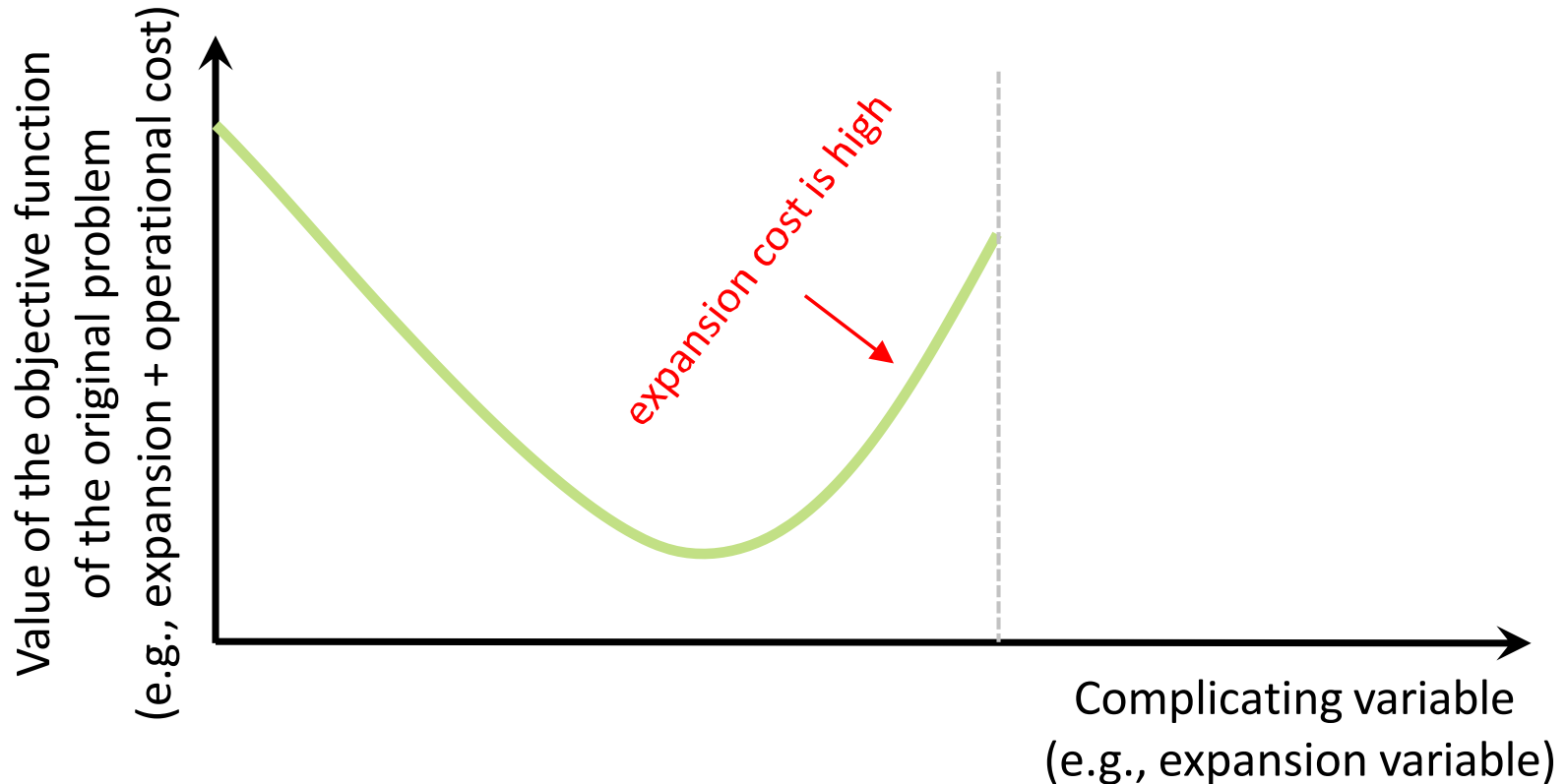
Concept-capacity expansion problem



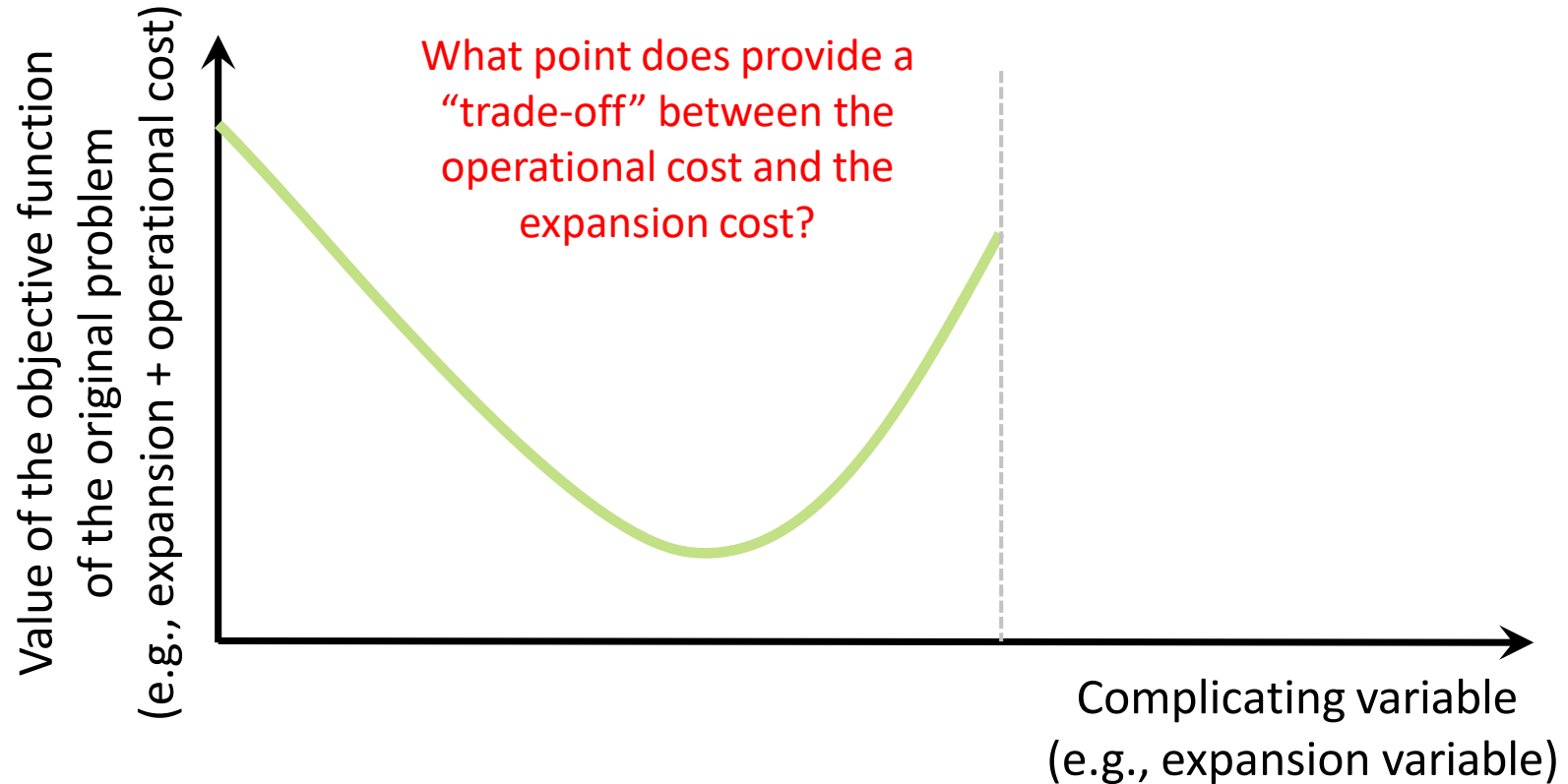
Concept-capacity expansion problem



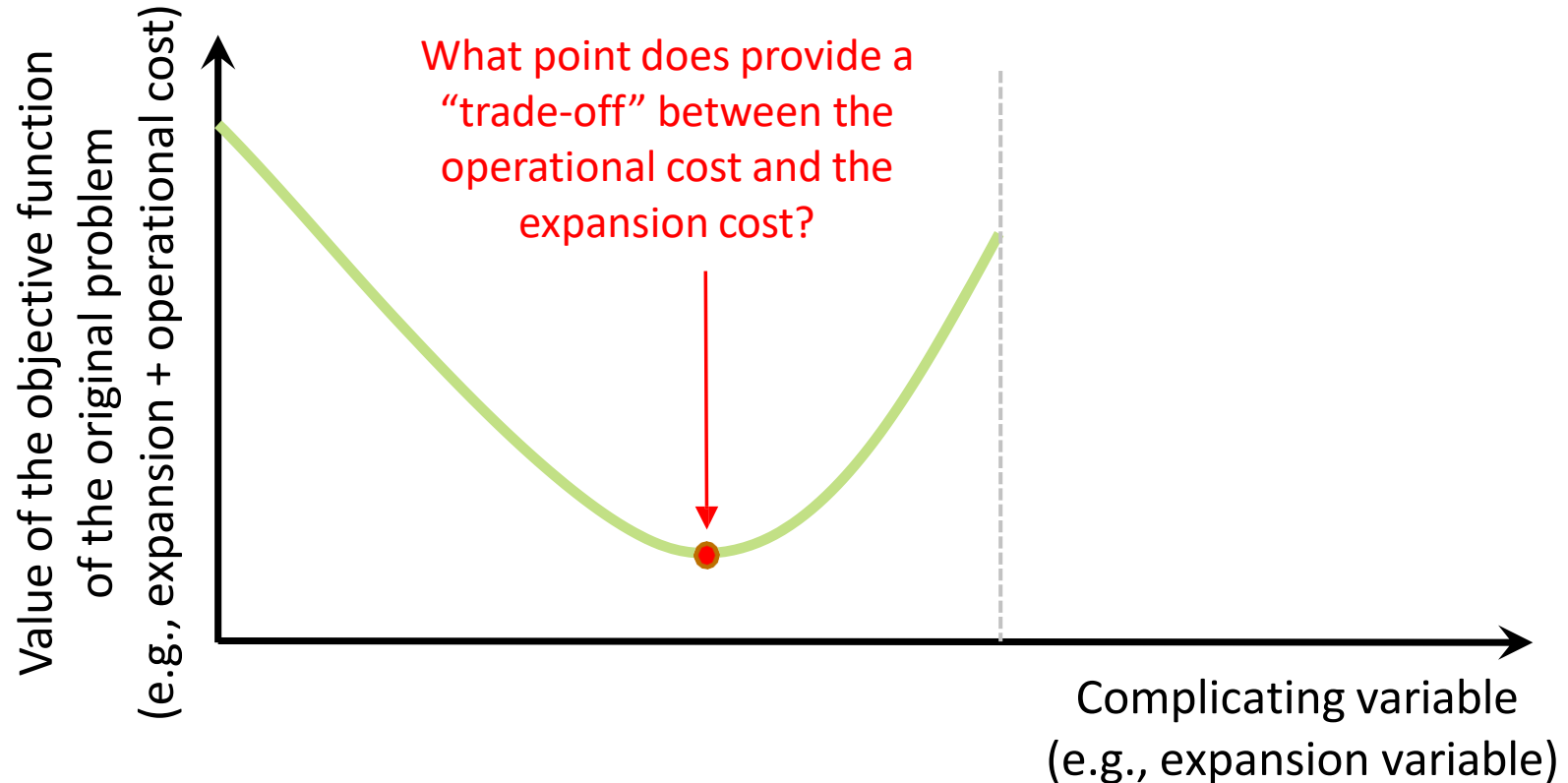
Concept-capacity expansion problem



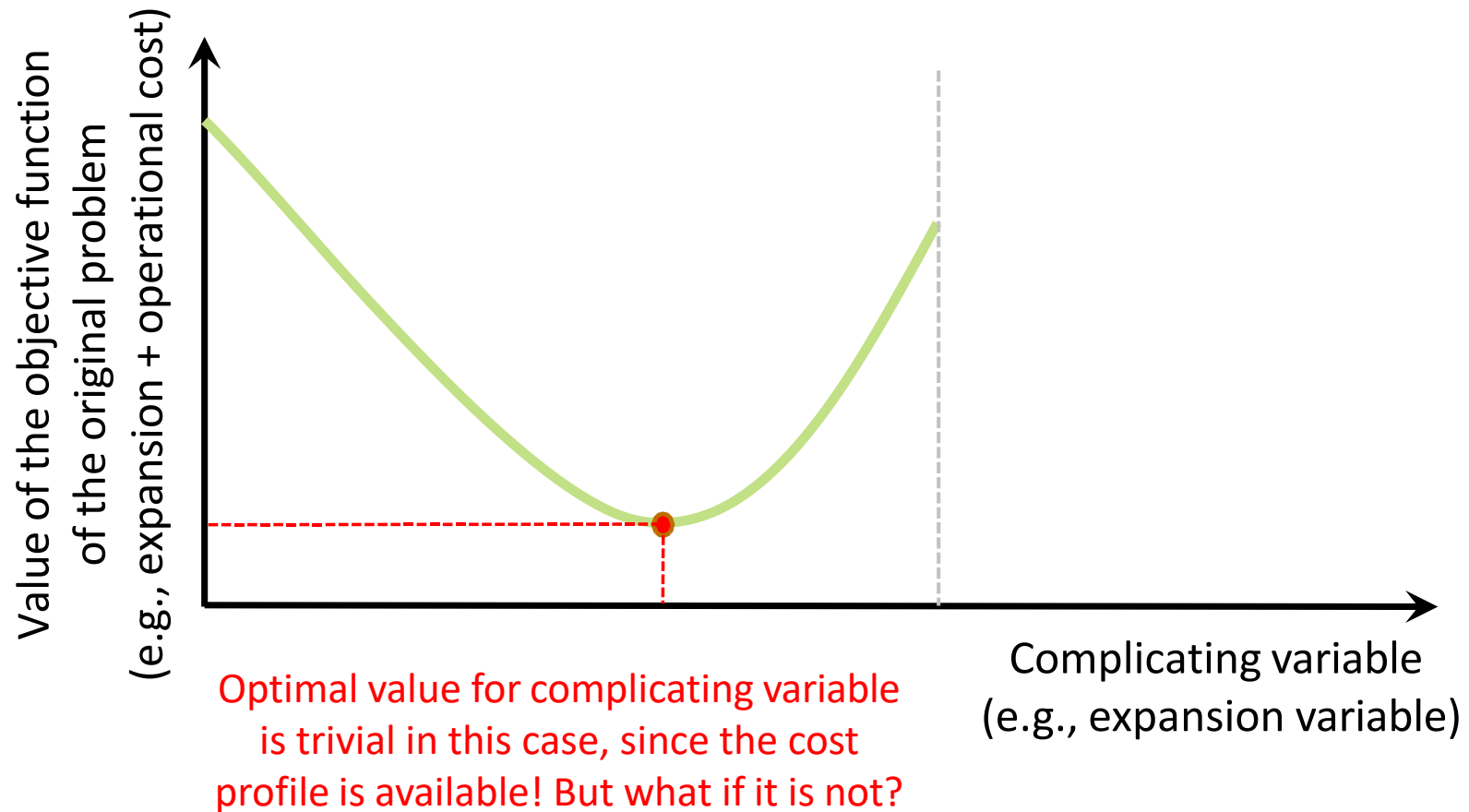
Concept-capacity expansion problem



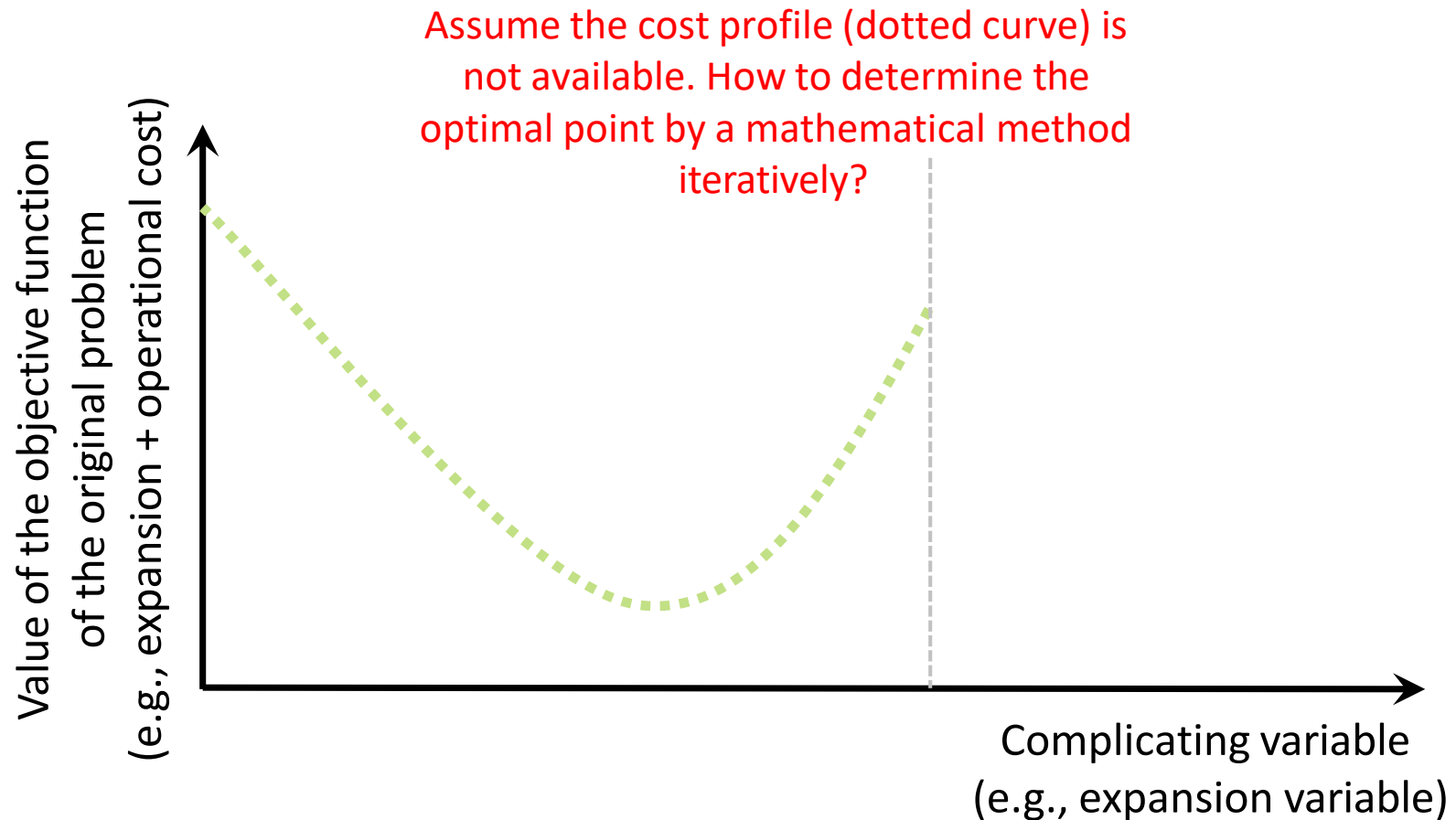
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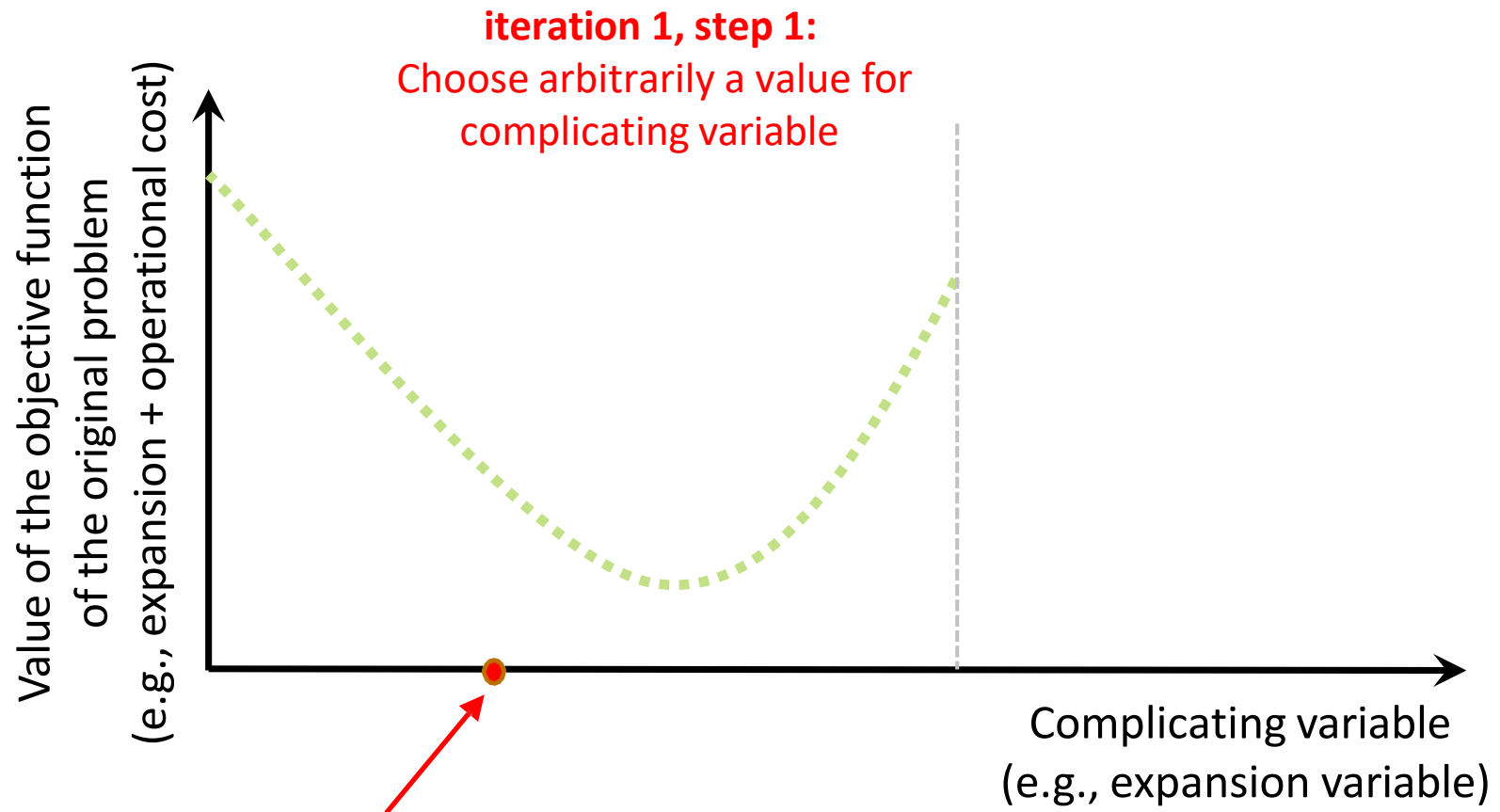
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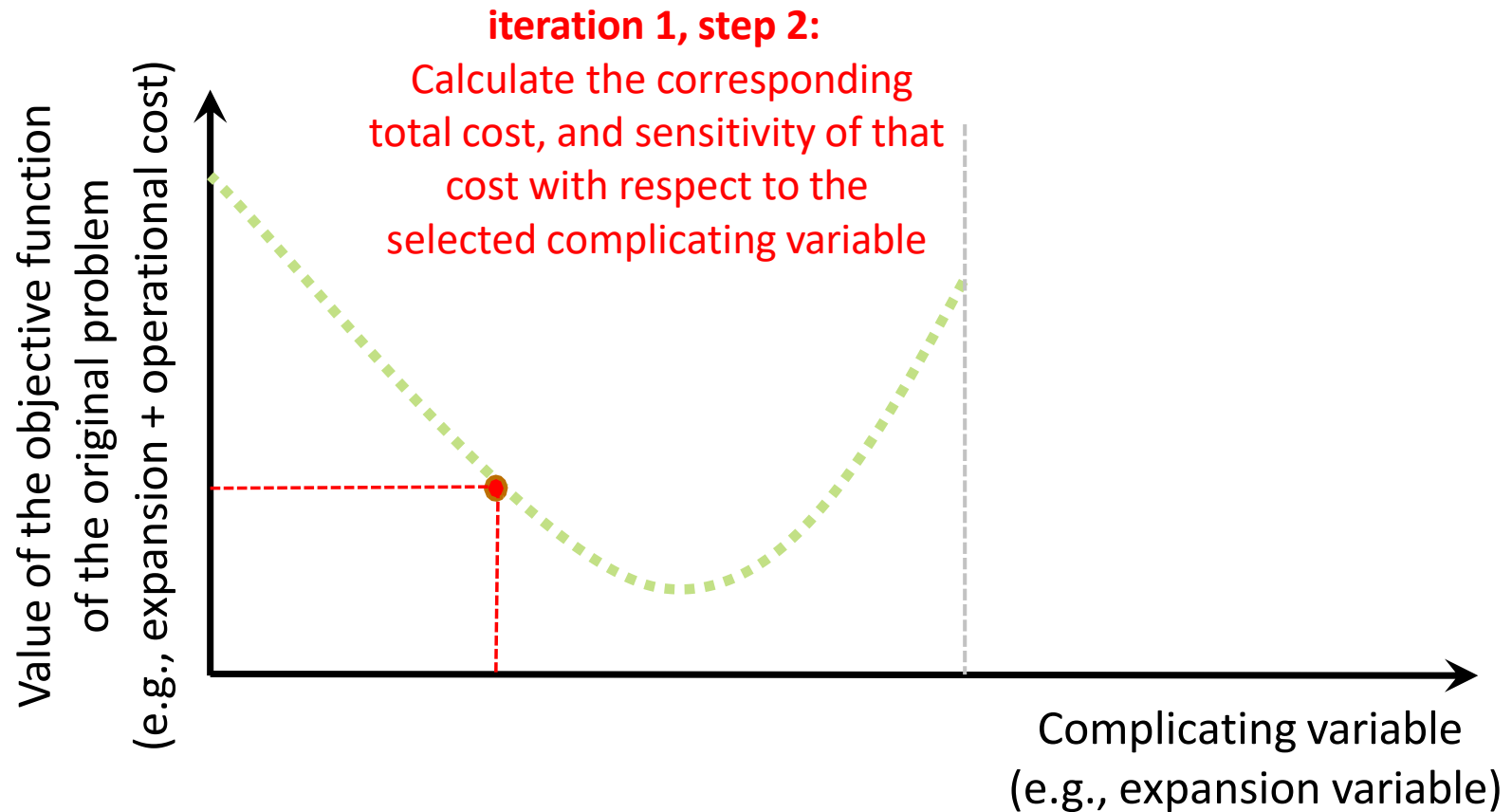
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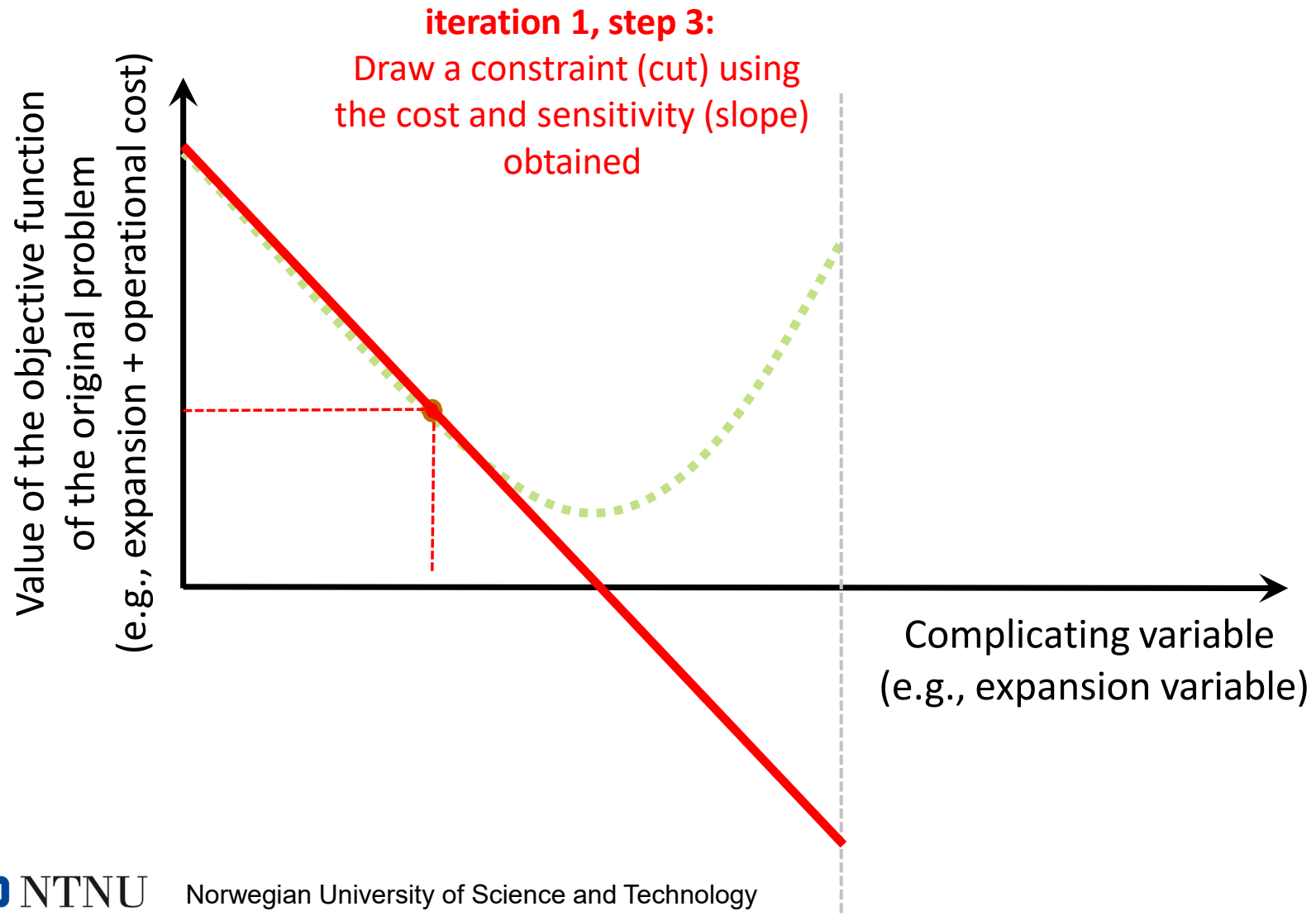
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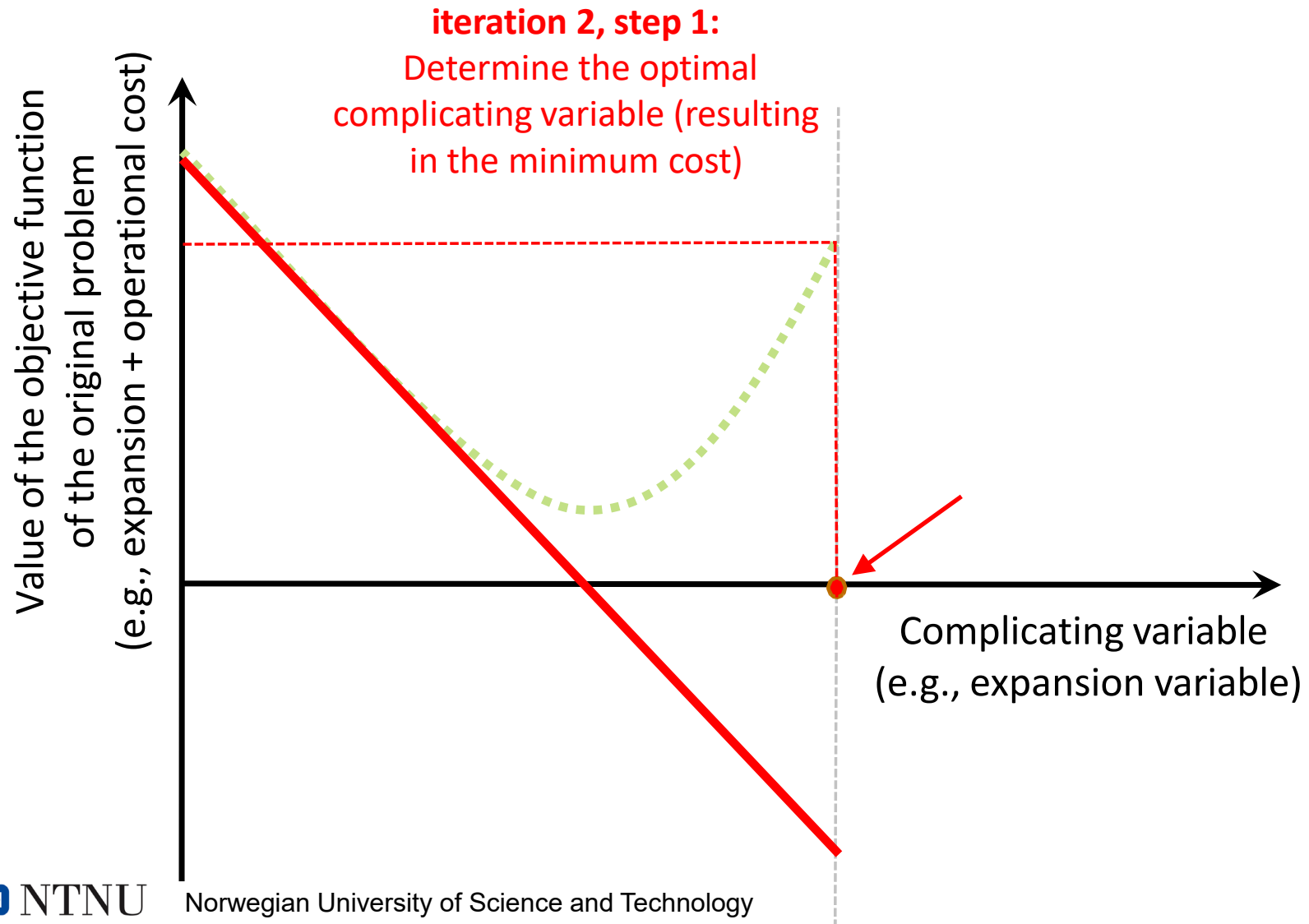
Concept-capacity expansion problem



Concept-capacity expansion problem

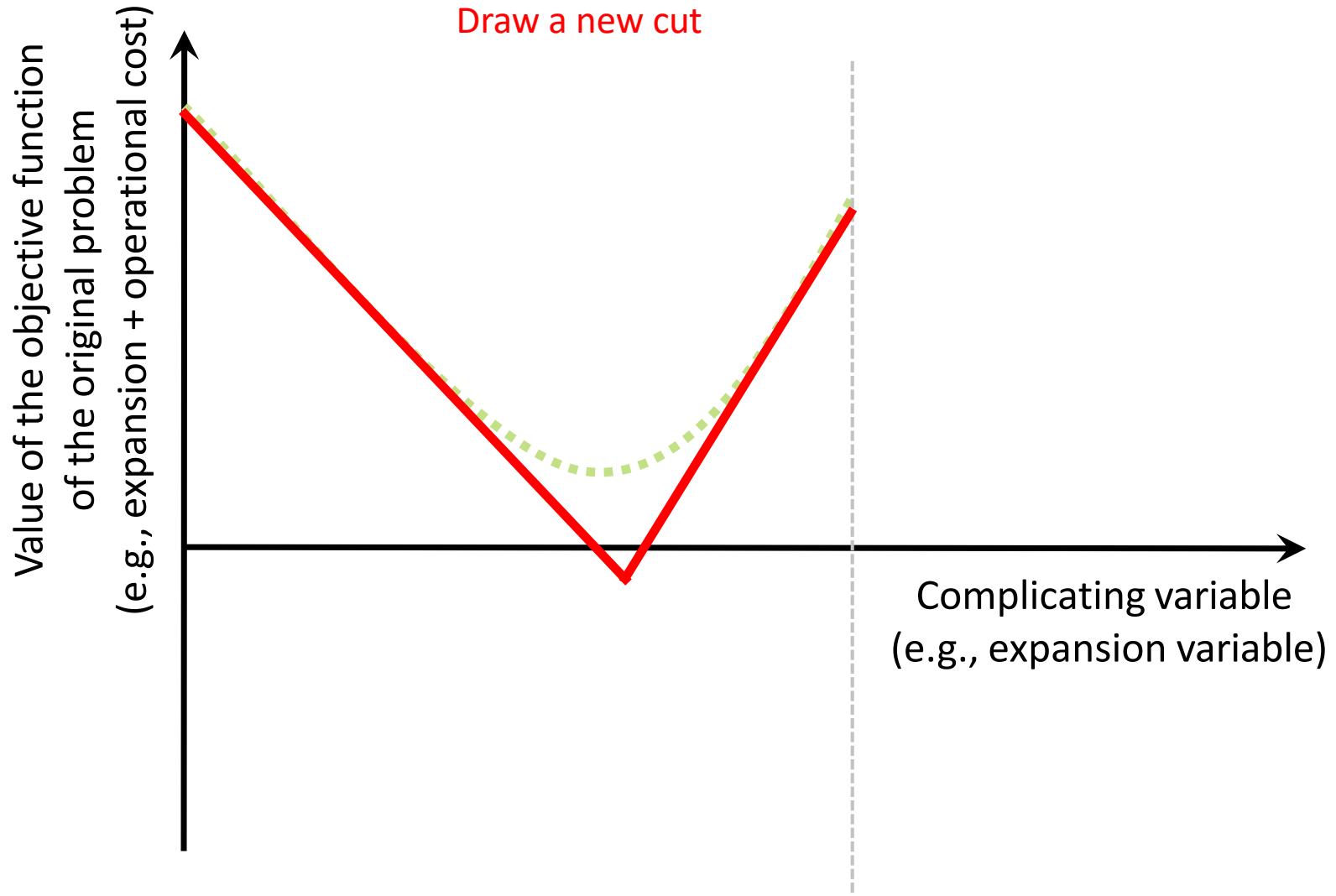


Concept-capacity expansion problem

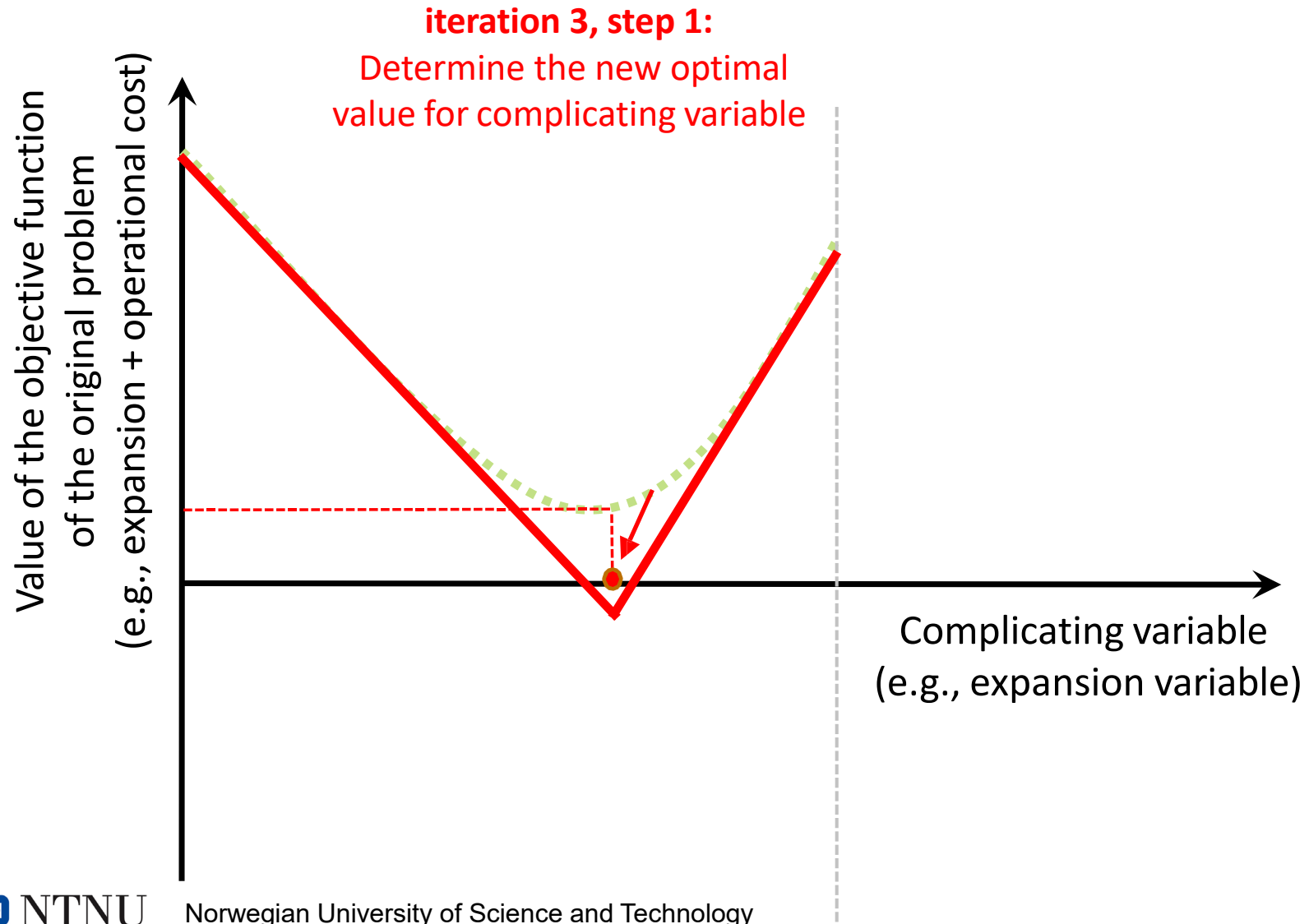


Concept-capacity expansion problem

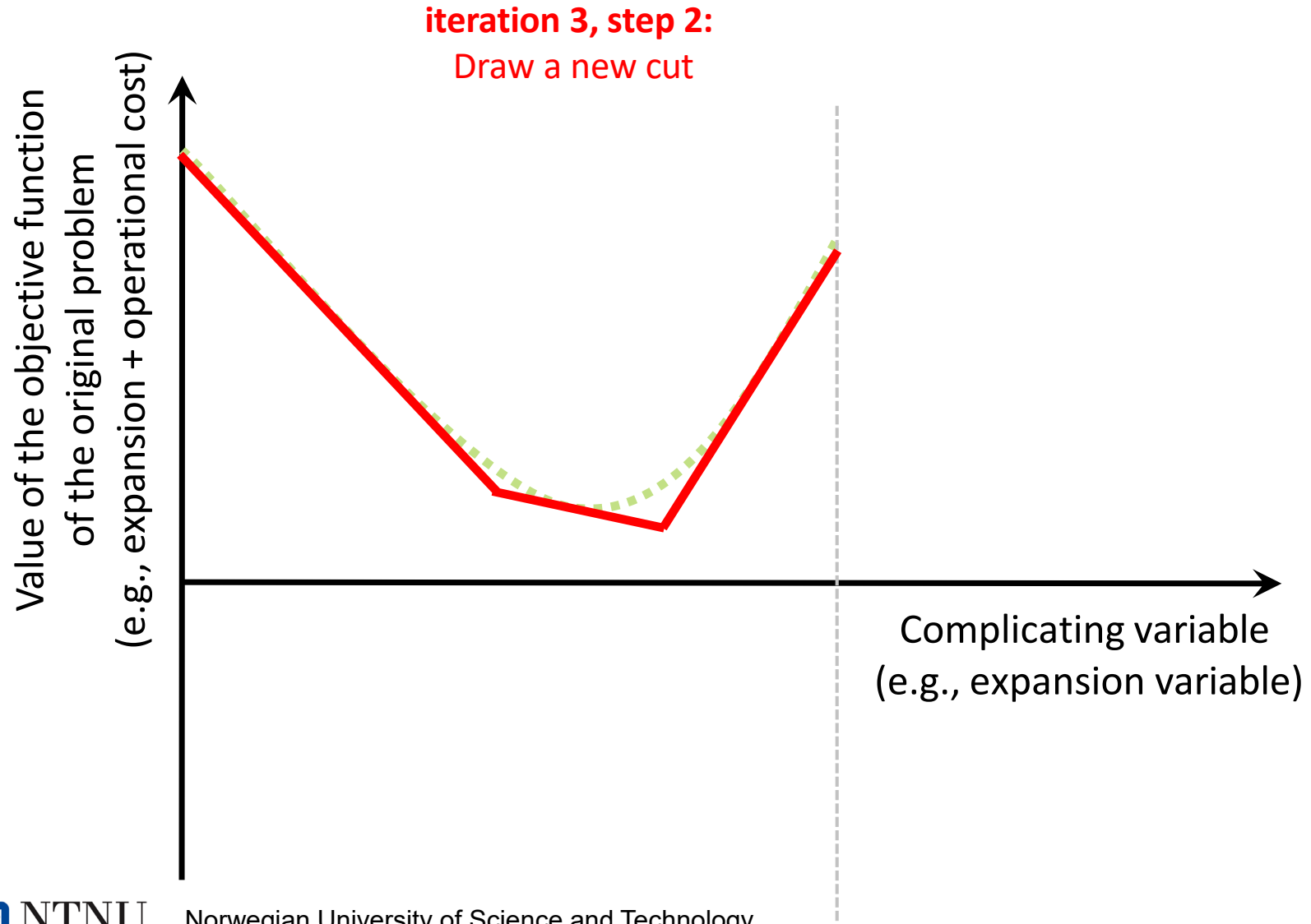
iteration 2, step 2:
Draw a new cut



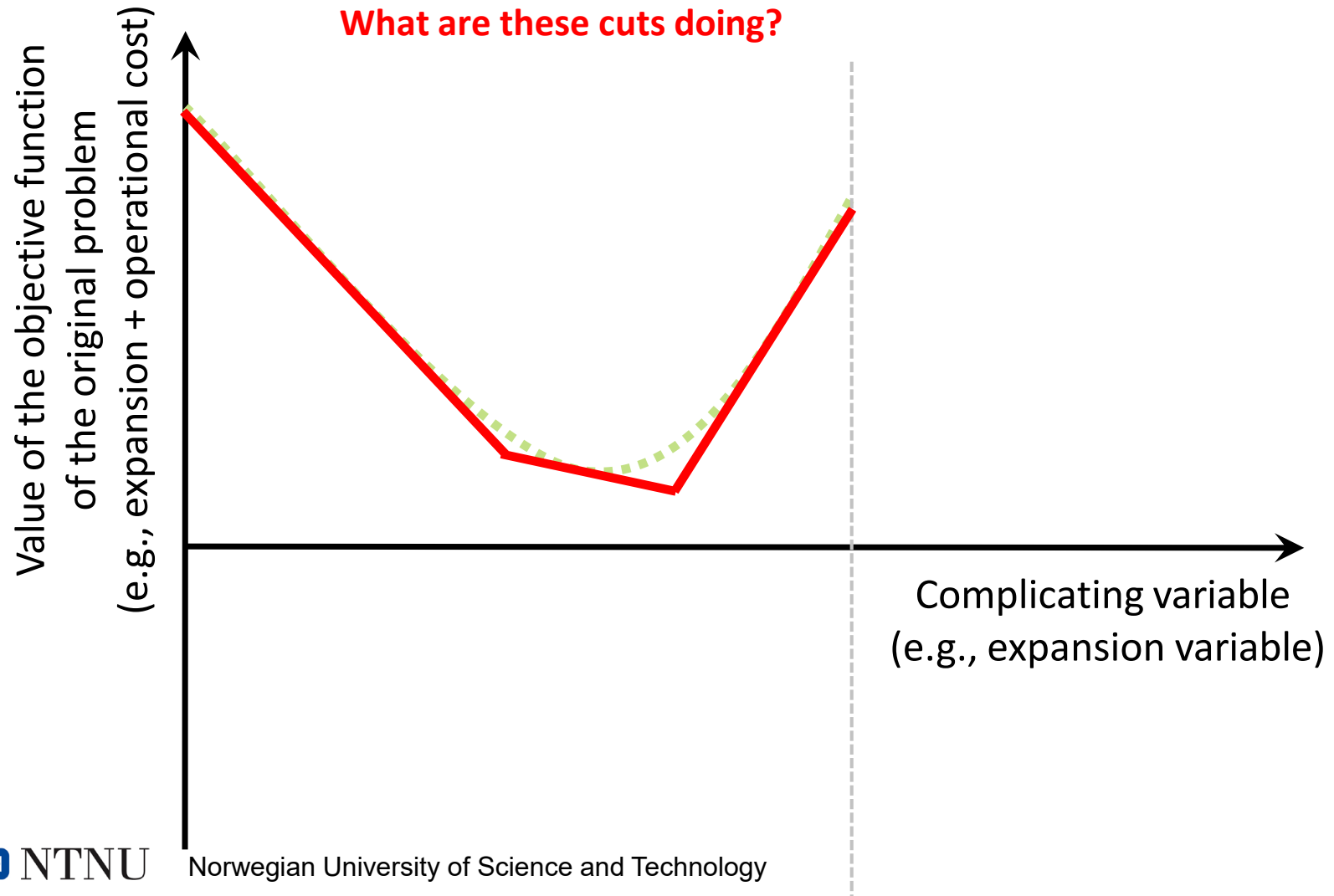
Concept-capacity expansion problem



Concept-capacity expansion problem

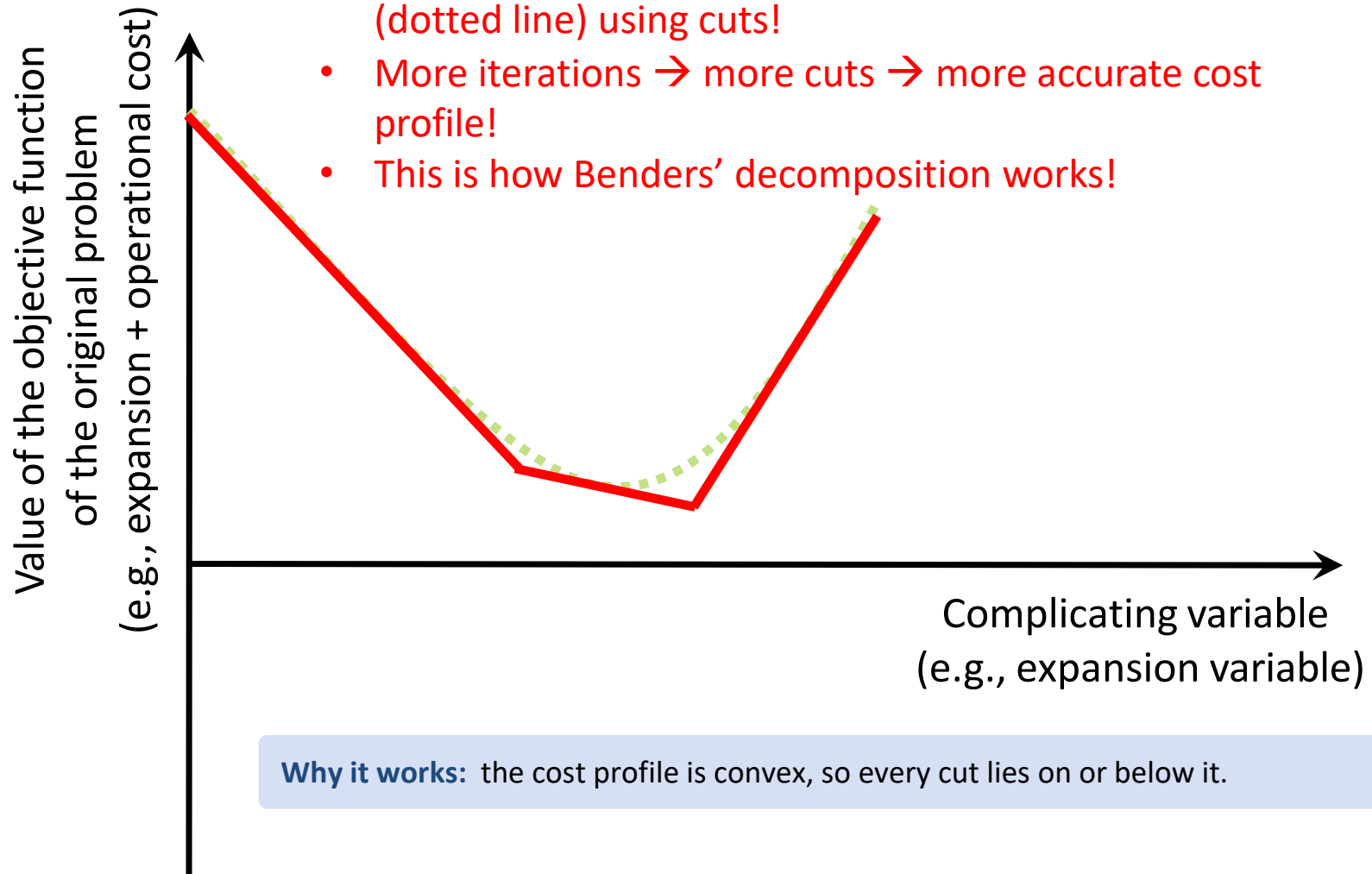


Concept-capacity expansion problem



Concept-capacity expansion problem

- We are indeed constructing the original cost profile (dotted line) using cuts!
- More iterations $\rightarrow$ more cuts $\rightarrow$ more accurate cost profile!
- This is how Benders' decomposition works!



Why it works: the cost profile is convex, so every cut lies on or below it.

From the picture to the algorithm

In the picture	In Benders' decomposition
Choose a value for the complicating variable	Fix the complicating variable to a trial value
Calculate the total cost and its sensitivity (slope)	Solve the subproblem: objective value and dual variables
Draw a cut through that point	Add a Benders' cut to the master problem
Find the lowest point under all the cuts	Solve the master problem: new trial value
Cost on the curve at the trial point	Upper bound (UB)
Lowest point of the cuts	Lower bound (LB)
The cuts reproduce the curve near the optimum	UB and LB meet: stop

Questions for this part

- How do we write the subproblem and the master problem?
- Where do the sensitivities (dual variables) enter?
- When do we stop iterating?

Illustrative Example-Book section 3.3

$$\begin{array}{ll}\text{minimise} & z = -y - \frac{x}{4} \\ \text{subject to,} & \end{array}$$

$$y - x \leq 5$$

$$y - \frac{1}{2}x \leq \frac{15}{2}$$

$$y + \frac{1}{2}x \leq \frac{32}{2}$$

$$-y + x \leq 10$$

$$0 \leq x \leq 16$$

$$y \geq 0$$

Let's consider x as a complicating variable, and assume x is fixed!

Illustrative Example

i : current Bender's iteration

k : set of previous iteration

- For given value for complicating variable (i.e., x), subproblem obtains the optimal value for y and dual variables (sensitivities).

Subproblem:

minimise $-y^{(i)}$

subject to,

$$y^{(i)} \leq 5 + x^{\text{fixed}(i)} \quad : \pi^{(i)}$$

$$y^{(i)} \leq \frac{15}{2} + \frac{x^{\text{fixed}(i)}}{2} \quad : \mu^{(i)}$$

$$y^{(i)} \leq \frac{32}{2} - \frac{x^{\text{fixed}(i)}}{2} \quad : \sigma^{(i)}$$

$$-y^{(i)} \leq 10 - x^{\text{fixed}(i)} \quad : \gamma^{(i)}$$

$$y^{(i)} \geq 0$$

Where does the Benders' cut come from?

1 Subproblem for a fixed x

minimise $-y$

subject to,

$$y \leq 5 + x \quad : \pi$$

$$y \leq \frac{15}{2} + \frac{x}{2} \quad : \mu$$

$$y \leq \frac{32}{2} - \frac{x}{2} \quad : \sigma$$

$$-y \leq 10 - x \quad : \gamma$$

$$y \geq 0$$

x appears only on the right-hand sides.

2 Its dual: duals $\times$ right-hand sides

$$\underset{\pi, \mu, \sigma, \gamma}{\text{maximise}} \left\{ -\pi[5 + x] - \mu\left[\frac{15}{2} + \frac{x}{2}\right] - \sigma\left[\frac{32}{2} - \frac{x}{2}\right] - \gamma[10 - x] \right\}$$

Subject to,

$$\pi + \mu + \sigma - \gamma \geq 1$$

$$\pi, \mu, \sigma, \gamma \geq 0$$

Same optimal value as the subproblem (strong duality), and no x in its constraints.

- 3 The duals from iteration k remain feasible in the dual for any x , so they give a lower estimate:
subproblem cost at $x \geq -\pi^k[5 + x] - \mu^k\left[\frac{15}{2} + x/2\right] - \sigma^k\left[\frac{32}{2} - x/2\right] - \gamma^k[10 - x]$
- 4 In the master problem α stands for that cost, so $\alpha \geq$ the same expression: **the Benders' cut**

Illustrative Example

i : current Bender's iteration

k : set of previous iteration

- For given value for complicating variable (i.e., x), subproblem obtains the optimal value for y and dual variables (sensitivities).
- Based on values for sensitivities achieved in subproblem, master problem obtains the updated value for complicating variable x .

Subproblem:

$$\begin{aligned} &\underset{y^{(i)}}{\text{minimise}} -y^{(i)} \\ &\text{subject to,} \\ &y^{(i)} \leq 5 + x^{\text{fixed}(i)} \quad : \pi^{(i)} \\ &y^{(i)} \leq \frac{15}{2} + \frac{x^{\text{fixed}(i)}}{2} \quad : \mu^{(i)} \\ &y^{(i)} \leq \frac{32}{2} - \frac{x^{\text{fixed}(i)}}{2} \quad : \sigma^{(i)} \\ &-y^{(i)} \leq 10 - x^{\text{fixed}(i)} \quad : \gamma^{(i)} \\ &y^{(i)} \geq 0 \end{aligned}$$

Master problem:

$$\underset{x^{(i)}, \alpha^{(i)}}{\text{minimise}} -\frac{x^{(i)}}{4} - \alpha^{(i)}$$

subject to,

$$0 \leq x^{(i)} \leq 16$$

$$\alpha^{(i)} \geq \alpha^{\text{down}}$$

$$\alpha^{(i)} \geq -\pi^{(k)} \left[5 + x^{(i)} \right] - \mu^{(k)} \left[\frac{15}{2} + \frac{x^{(i)}}{2} \right] - \sigma^{(k)} \left[\frac{32}{2} - \frac{x^{(i)}}{2} \right] - \gamma^{(k)} \left[10 - x^{(i)} \right] \quad \forall k = 1, \dots, i-1$$

This is an auxiliary variable, representing the objective function of the subproblem(s) within the objective function of the original problem!

Illustrative Example

i : current Bender's iteration

k : set of previous iteration

- For given value for complicating variable (i.e., x), subproblem obtains the optimal value for y and dual variables (sensitivities).
- Based on values for sensitivities achieved in subproblem, master problem obtains the updated value for complicating variable x .

Subproblem:

$$\begin{aligned} &\underset{y^{(i)}}{\text{minimise}} -y^{(i)} \\ &\text{subject to,} \\ &y^{(i)} \leq 5 + x^{\text{fixed}(i)} \quad : \pi^{(i)} \\ &y^{(i)} \leq \frac{15}{2} + \frac{x^{\text{fixed}(i)}}{2} \quad : \mu^{(i)} \\ &y^{(i)} \leq \frac{32}{2} - \frac{x^{\text{fixed}(i)}}{2} \quad : \sigma^{(i)} \\ &-y^{(i)} \leq 10 - x^{\text{fixed}(i)} \quad : \gamma^{(i)} \\ &y^{(i)} \geq 0 \end{aligned}$$

Master problem:

$$\underset{x^{(i)}, \alpha^{(i)}}{\text{minimise}} -\frac{x^{(i)}}{4} + \alpha^{(i)}$$

subject to,

$$0 \leq x^{(i)} \leq 16$$

$$\alpha^{(i)} \geq \alpha^{\text{down}}$$

To avoid unbounded solution in the first iteration when there is no cut (last constraint). It should be a very large negative constant, e.g., -10^6 .

$$\alpha^{(i)} \geq -\pi^{(k)} \left[5 + x^{(i)} \right] - \mu^{(k)} \left[\frac{15}{2} + \frac{x^{(i)}}{2} \right] - \sigma^{(k)} \left[\frac{32}{2} - \frac{x^{(i)}}{2} \right] - \gamma^{(k)} \left[10 - x^{(i)} \right] \quad \forall k = 1, \dots, i-1$$

Illustrative Example

i : current Bender's iteration

k : set of previous iteration

- For given value for complicating variable (i.e., x), subproblem obtains the optimal value for y and dual variables (sensitivities).
- Based on values for sensitivities achieved in subproblem, **master problem** obtains the updated value for complicating variable x .

Master problem:

$$\underset{x^{(i)}, \alpha^{(i)}}{\text{minimise}} -\frac{x^{(i)}}{4} + \alpha^{(i)}$$

subject to,

$$0 \leq x^{(i)} \leq 16$$

$$\alpha^{(i)} \geq \alpha^{\text{down}}$$

$$\alpha^{(i)} \geq -\pi^{(k)} \left[5 + x^{(i)} \right] - \mu^{(k)} \left[\frac{15}{2} + \frac{x^{(i)}}{2} \right] - \sigma^{(k)} \left[\frac{32}{2} - \frac{x^{(i)}}{2} \right] - \gamma^{(k)} \left[10 - x^{(i)} \right] \quad \forall k = 1, \dots, i-1$$

- The right hand side is the objective function of the **dual** subproblem
- This constraint generates "Benders' (optimality) cuts". One additional cut in each iteration!

Subproblem:

$$\underset{y^{(i)}}{\text{minimise}} -y^{(i)}$$

subject to,

$$y^{(i)} \leq 5 + x^{\text{fixed}(i)} \quad : \pi^{(i)}$$

$$y^{(i)} \leq \frac{15}{2} + \frac{x^{\text{fixed}(i)}}{2} \quad : \mu^{(i)}$$

$$y^{(i)} \leq \frac{32}{2} - \frac{x^{\text{fixed}(i)}}{2} \quad : \sigma^{(i)}$$

$$-y^{(i)} \leq 10 - x^{\text{fixed}(i)} \quad : \gamma^{(i)}$$

$$y^{(i)} \geq 0$$

Algorithm

- **Step 0: Initialization**

Set $i = 1$, $x^{\text{fixed}(1)} = x^{\text{initial}}$, and lower bound (LB) $= -\infty$

- **Step 1: Solve subproblem(s):** obtain the values of all dual variables (sensitivities), and the value of objective function, which is upper bound (UB)

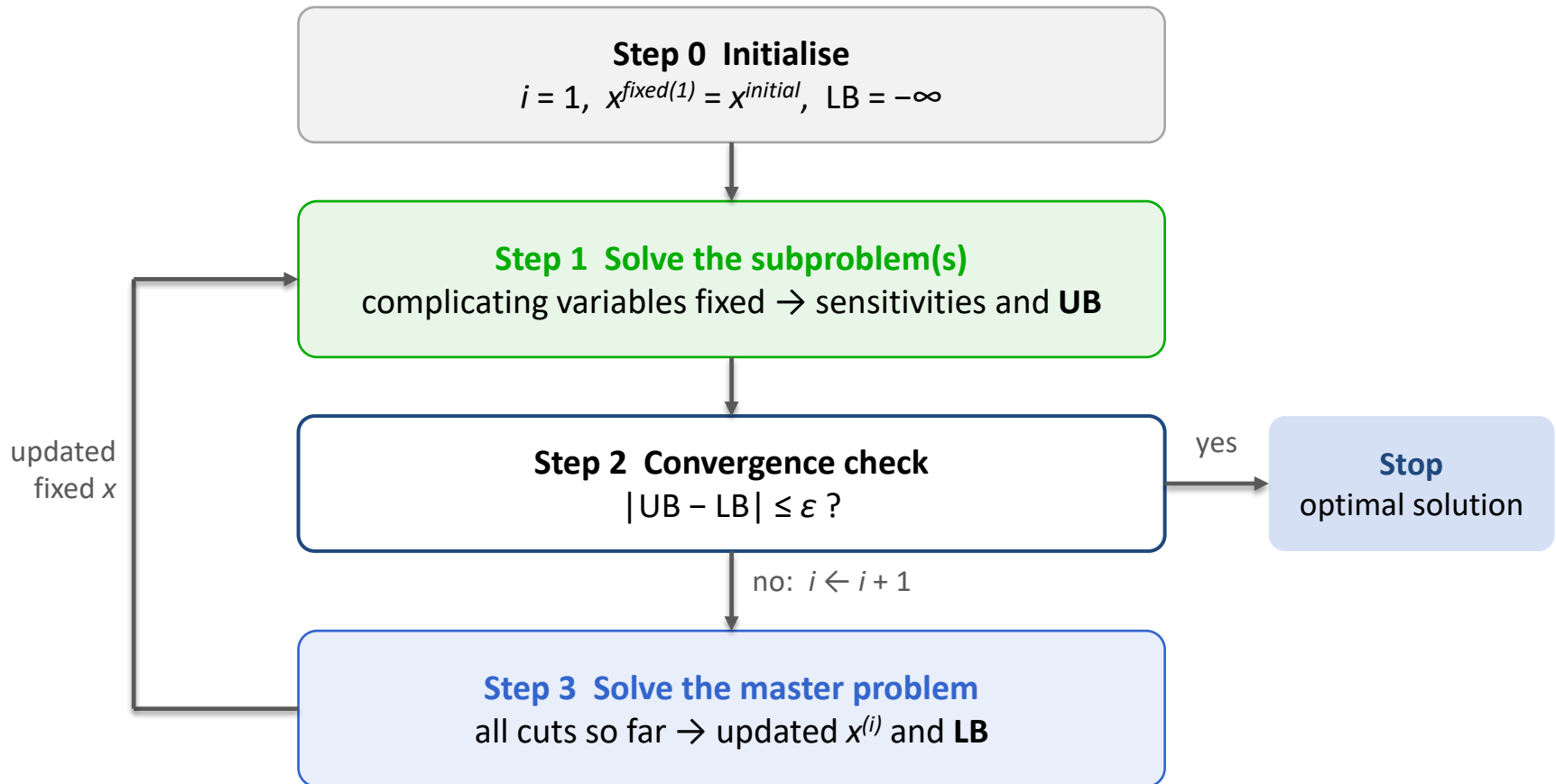
- **Step 2: Convergence check**

if $|UB - LB| \leq \epsilon$, then the optimal solution with a level of accuracy ϵ is obtained, otherwise $i \leftarrow i + 1$

- **Step 3: Solve master problem**

obtain the updated $x^{(i)}$ and the value of $\alpha^{(i)}$ as the updated LB, and go Step 1 with the updated fixed x

Algorithm at a glance



Remember: the subproblem gives the upper bound, the master problem the lower bound.

A compact form

Compare with the subproblem on the previous slides: what is new here, and what does it buy us?

Subproblem:

$$\underset{x^{(i)}, y^{(i)}}{\text{minimise}} -y^{(i)}$$

subject to,

$$y^{(i)} - x^{(i)} \leq 5 \quad : \pi^{(i)}$$

$$y^{(i)} - \frac{x^{(i)}}{2} \leq \frac{15}{2} \quad : \mu^{(i)}$$

$$y^{(i)} + \frac{x^{(i)}}{2} \leq \frac{32}{2} \quad : \sigma^{(i)}$$

$$-y^{(i)} + x^{(i)} \leq 10 \quad : \gamma^{(i)}$$

$$y^{(i)} \geq 0$$

$$x^{(i)} = x^{\text{fixed}(i)} \quad : \rho^{(i)}$$

A compact form

In this compact form, there is a constraint for fixing each complicating variable to a given value! However, x is now a “variable” in subproblem, though its value is fixed! Only one sensitivity ($\rho^{(i)}$) is needed in the master problem! Indeed, it is the sensitivity/subgradient of the subproblem value function with respect to x

Subproblem:

$$\underset{x^{(i)}, y^{(i)}}{\text{minimise}} -y^{(i)}$$

subject to,

$$y^{(i)} - x^{(i)} \leq 5 \quad : \pi^{(i)}$$

$$y^{(i)} - \frac{x^{(i)}}{2} \leq \frac{15}{2} \quad : \mu^{(i)}$$

$$y^{(i)} + \frac{x^{(i)}}{2} \leq \frac{32}{2} \quad : \sigma^{(i)}$$

$$-y^{(i)} + x^{(i)} \leq 10 \quad : \gamma^{(i)}$$

$$y^{(i)} \geq 0$$

$$x^{(i)} = x^{\text{fixed}(i)} \quad : \rho^{(i)}$$

A compact form

Subproblem:

$$\begin{aligned} & \underset{x^{(i)}, y^{(i)}}{\text{minimise}} && -y^{(i)} \\ & \text{subject to,} && \\ & y^{(i)} - x^{(i)} \leq 5 && : \pi^{(i)} \\ & y^{(i)} - \frac{x^{(i)}}{2} \leq \frac{15}{2} && : \mu^{(i)} \\ & y^{(i)} + \frac{x^{(i)}}{2} \leq \frac{32}{2} && : \sigma^{(i)} \\ & -y^{(i)} + x^{(i)} \leq 10 && : \gamma^{(i)} \\ & y^{(i)} \geq 0 && \\ & x^{(i)} = x^{\text{fixed}(i)} && : \rho^{(i)} \end{aligned}$$

Master problem:

$$\begin{aligned} & \underset{x^{(i)}, \alpha^{(i)}}{\text{minimise}} && -\frac{x^{(i)}}{4} + \alpha^{(i)} \\ & \text{subject to} && 0 \leq x^{(i)} \leq 16, \\ & && \alpha^{(i)} \geq \alpha^{\text{down}}, \\ & && \alpha^{(i)} \geq -y^{(k)} + \rho^{(k)}(x^{(i)} - x^{(k)}), \quad \forall k = 1, \dots, i-1. \end{aligned}$$

A compact form

Same cut, written differently

slope: $\rho^{(k)} = -\pi^{(k)} - \frac{\mu^{(k)}}{2} + \frac{\sigma^{(k)}}{2} + \gamma^{(k)}$

value at $x^{(k)}$: $-y^{(k)}$ (strong duality)

ρ is a sensitivity (either sign); π, μ, σ, γ are multipliers ≥ 0 .

Check: Equivalence of Extended and Compact Benders' Cuts in PMO - Reading materials

Master problem:

$$\underset{x^{(i)}, \alpha^{(i)}}{\text{minimise}} \quad -\frac{x^{(i)}}{4} + \alpha^{(i)}$$

$$\text{subject to} \quad 0 \leq x^{(i)} \leq 16,$$

$$\alpha^{(i)} \geq \alpha^{\text{down}},$$

$$\alpha^{(i)} \geq -y^{(k)} + \rho^{(k)}(x^{(i)} - x^{(k)}), \quad \forall k = 1, \dots, i-1.$$

A compact formulation for Benders' cut!



Subproblem:

$$\underset{x^{(i)}, y^{(i)}}{\text{minimise}} \quad -y^{(i)}$$

subject to,

$$y^{(i)} - x^{(i)} \leq 5 \quad : \pi^{(i)}$$

$$y^{(i)} - \frac{x^{(i)}}{2} \leq \frac{15}{2} \quad : \mu^{(i)}$$

$$y^{(i)} + \frac{x^{(i)}}{2} \leq \frac{32}{2} \quad : \sigma^{(i)}$$

$$-y^{(i)} + x^{(i)} \leq 10 \quad : \gamma^{(i)}$$

$$y^{(i)} \geq 0$$

$$x^{(i)} = x^{\text{fixed}(i)} \quad : \rho^{(i)}$$

Illustrative example: the iterations

Start from $x^{(1)} = 0$ and follow the algorithm. UB = $-x/4 - y$ (subproblem), LB = $-x/4 + \alpha$ (master problem).

Iteration i	$x^{(i)}$ (trial value)	LB	$y^{(i)}$	sensitivity $\rho^{(i)}$	UB
1	0 (initial)	$-\infty$	5	-1	-5
2	16	-25	8	+0.5	-12
3	7.33	-14.17	11.17	-0.5	-13
4	8.5	-13.875	11.75	-	-13.875

Cuts added to the master problem

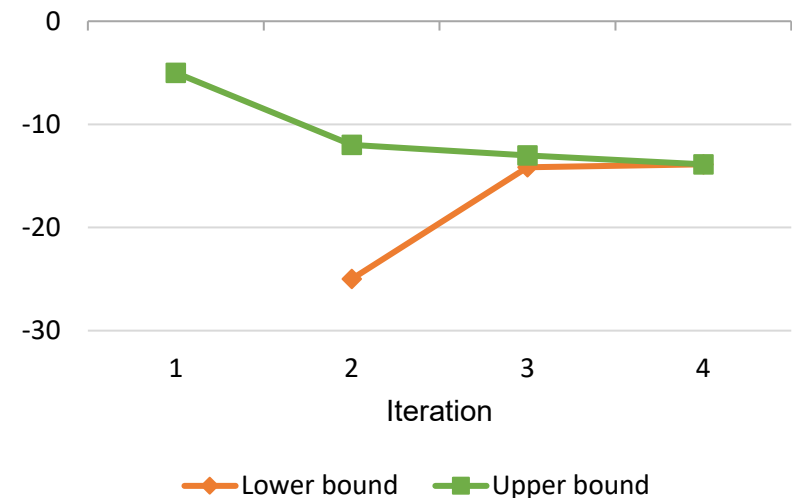
after iteration 1: $\alpha \geq -5 - x$ $\pi = 1, \rho = -1$

after iteration 2: $\alpha \geq -16 + 0.5x$ $\sigma = 1, \rho = +0.5$

after iteration 3: $\alpha \geq -7.5 - 0.5x$ $\mu = 1, \rho = -0.5$

Iteration 4: UB = LB = -13.875, so we stop.

Optimum: $x^* = 8.5, y^* = 11.75, z^* = -13.875$



Applications of Benders' decomposition to two-stage scenario-based stochastic programming

Two-stage scenario-based stochastic programming

First stage cost



$$\text{Minimize}_{x, y_\omega} \boxed{f(x)} + \sum_{\omega} \pi_{\omega} g(y_{\omega})$$

subject to

$$h(x) \leq a$$

$$k(x) + l(y_{\omega}) \leq b_{\omega} \quad \forall \omega$$

Two-stage scenario-based stochastic programming

Minimize $f(x) + \sum_{\omega} \pi_{\omega} g(y_{\omega})$

subject to

$$h(x) \leq a$$
$$k(x) + l(y_{\omega}) \leq b_{\omega} \quad \forall \omega$$



Probability of scenario w

Notation: from here on, π_{ω} is the probability of scenario ω , not the multiplier π of the illustrative example.

Two-stage scenario-based stochastic programming

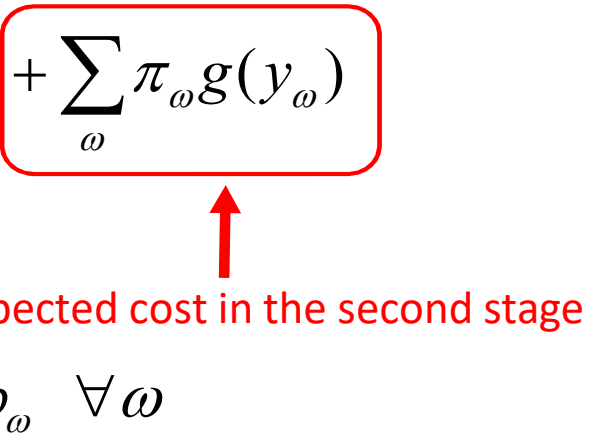
Minimize $f(x) + \sum_{\omega} \pi_{\omega} g(y_{\omega})$

subject to

$h(x) \leq a$

$k(x) + l(y_{\omega}) \leq b_{\omega} \quad \forall \omega$

Expected cost in the second stage



Two-stage scenario-based stochastic programming

$$\underset{x, y_\omega}{\text{Minimize}} \quad f(x) + \sum_{\omega} \pi_{\omega} g(y_{\omega})$$

subject to

$$h(x) \leq a$$



First-stage constraint

$$k(x) + l(y_{\omega}) \leq b_{\omega} \quad \forall \omega$$



Linking constraint

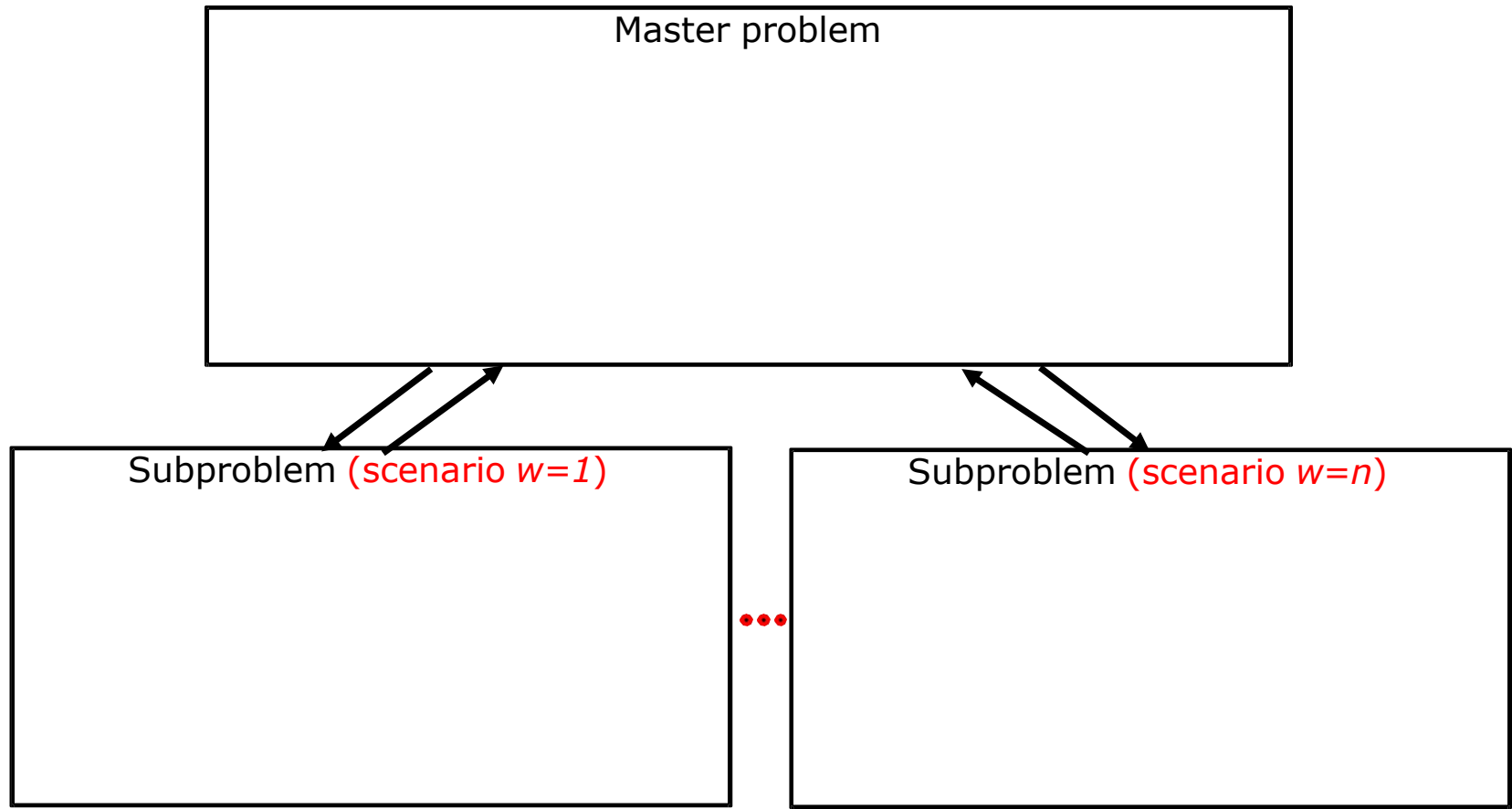
Two-stage scenario-based stochastic programming

$$\begin{aligned} & \underset{x, y_\omega}{\text{Minimize}} && f(x) + \sum_{\omega} \pi_{\omega} g(y_{\omega}) \\ & \text{subject to} && \\ & && h(x) \leq a \\ & && k(x) + l(y_{\omega}) \leq b_{\omega} \quad \forall \omega \end{aligned}$$

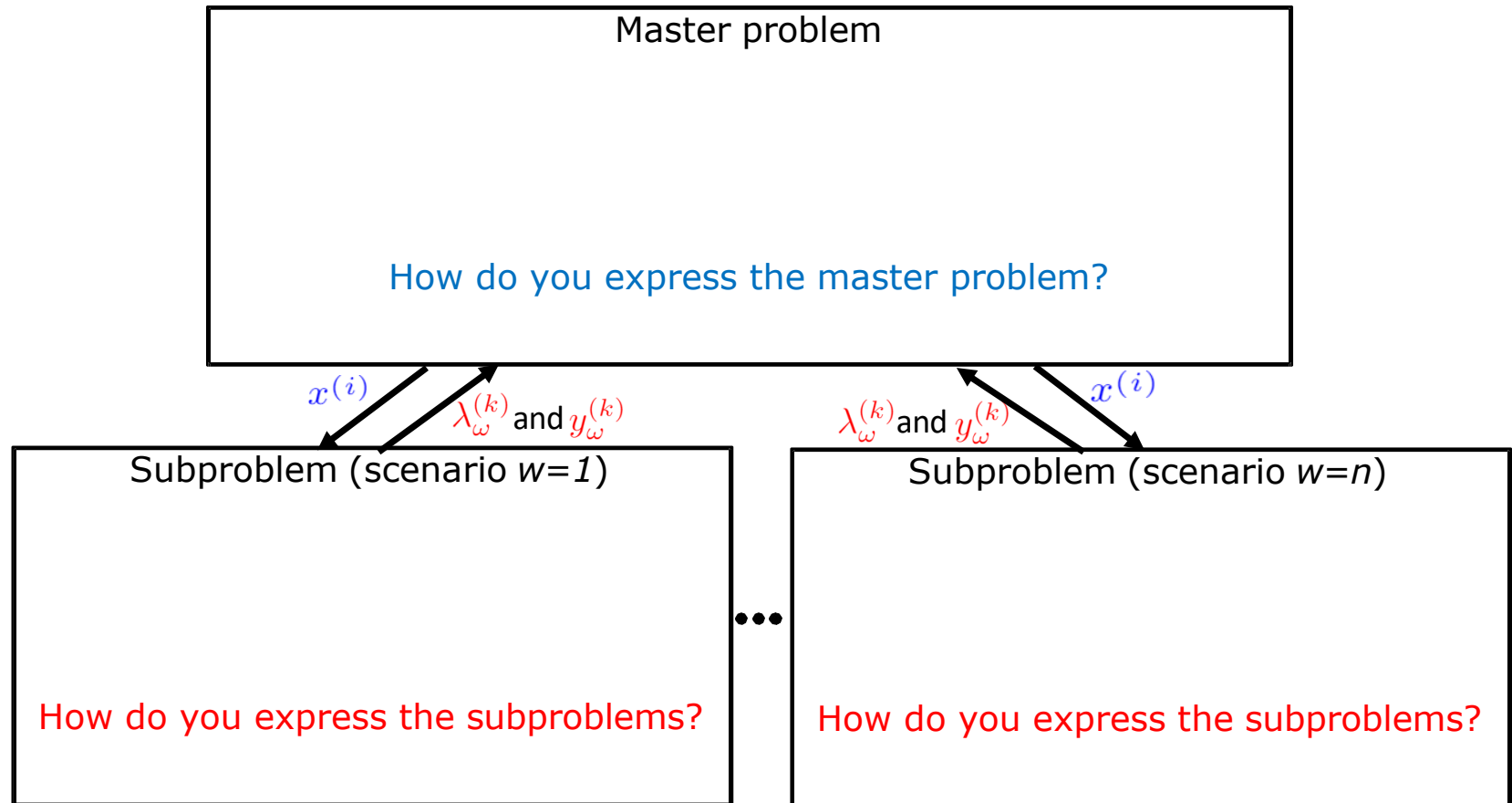
Recall that the first-stage variables (vector $\mathbf{x}$) are complicating variables!

- We need to formulate:
 - One master problem
 - One subproblem per scenario

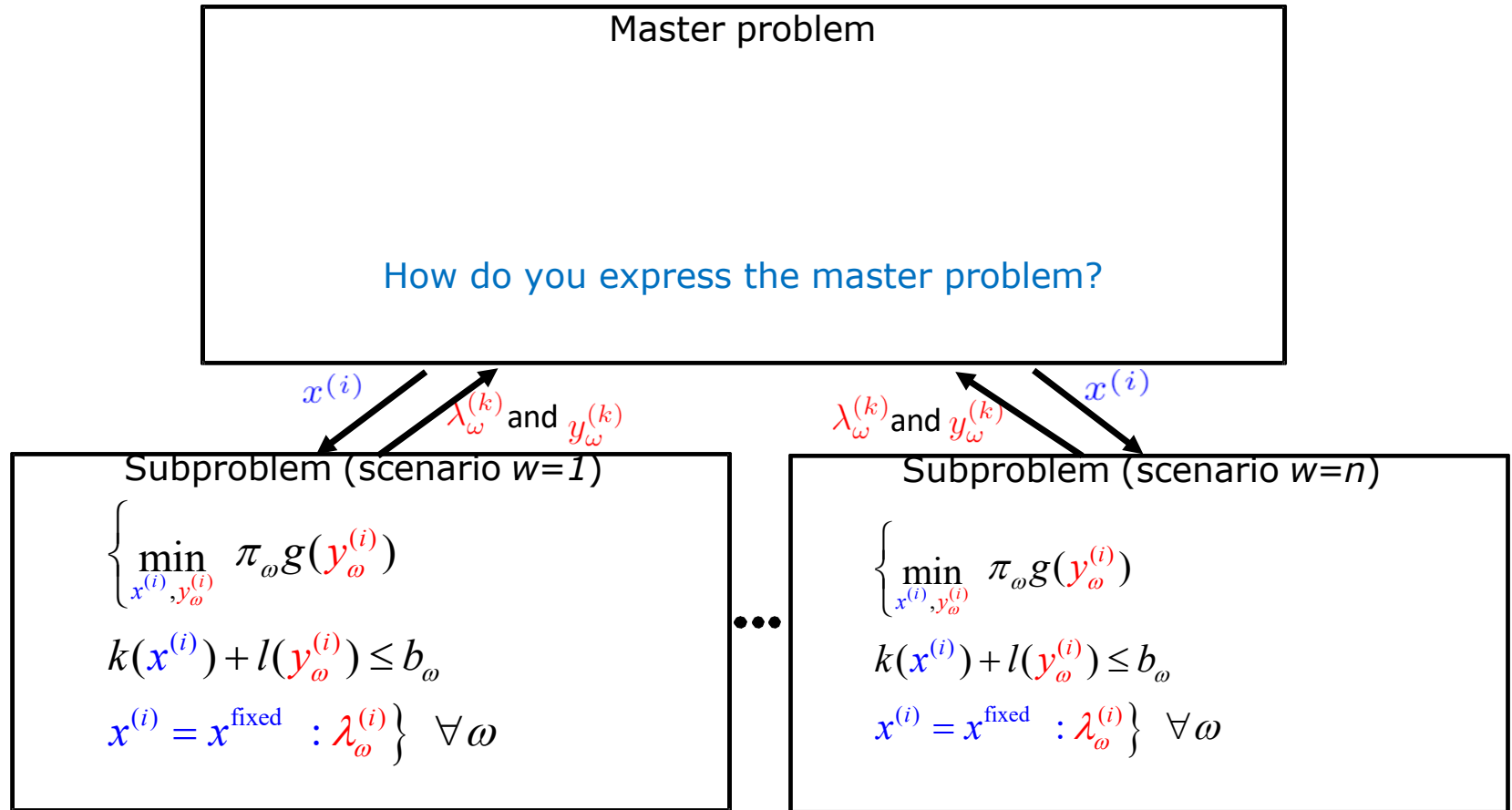
Two-stage stochastic program: Benders' solution



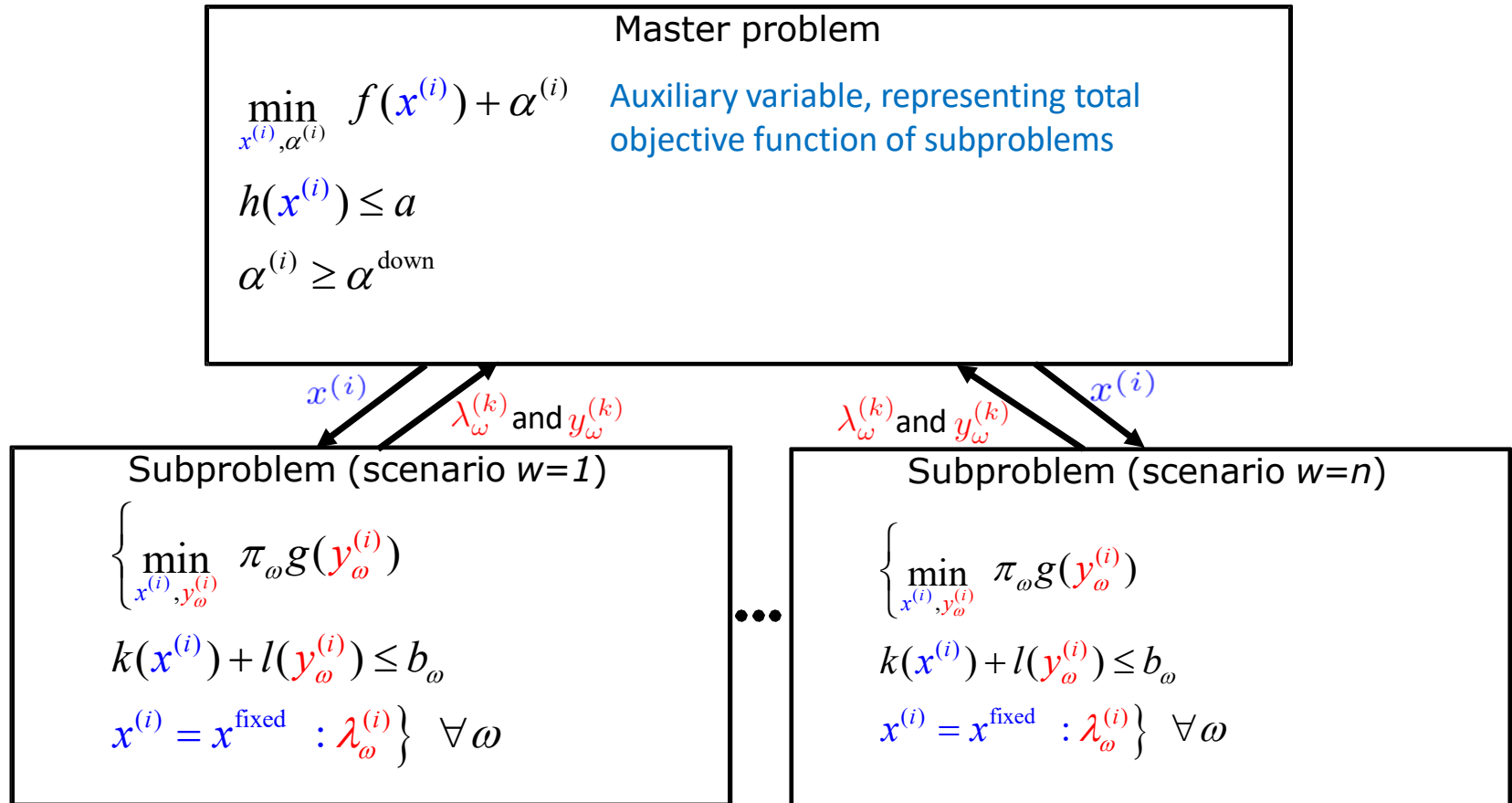
Two-stage stochastic program: Benders' solution



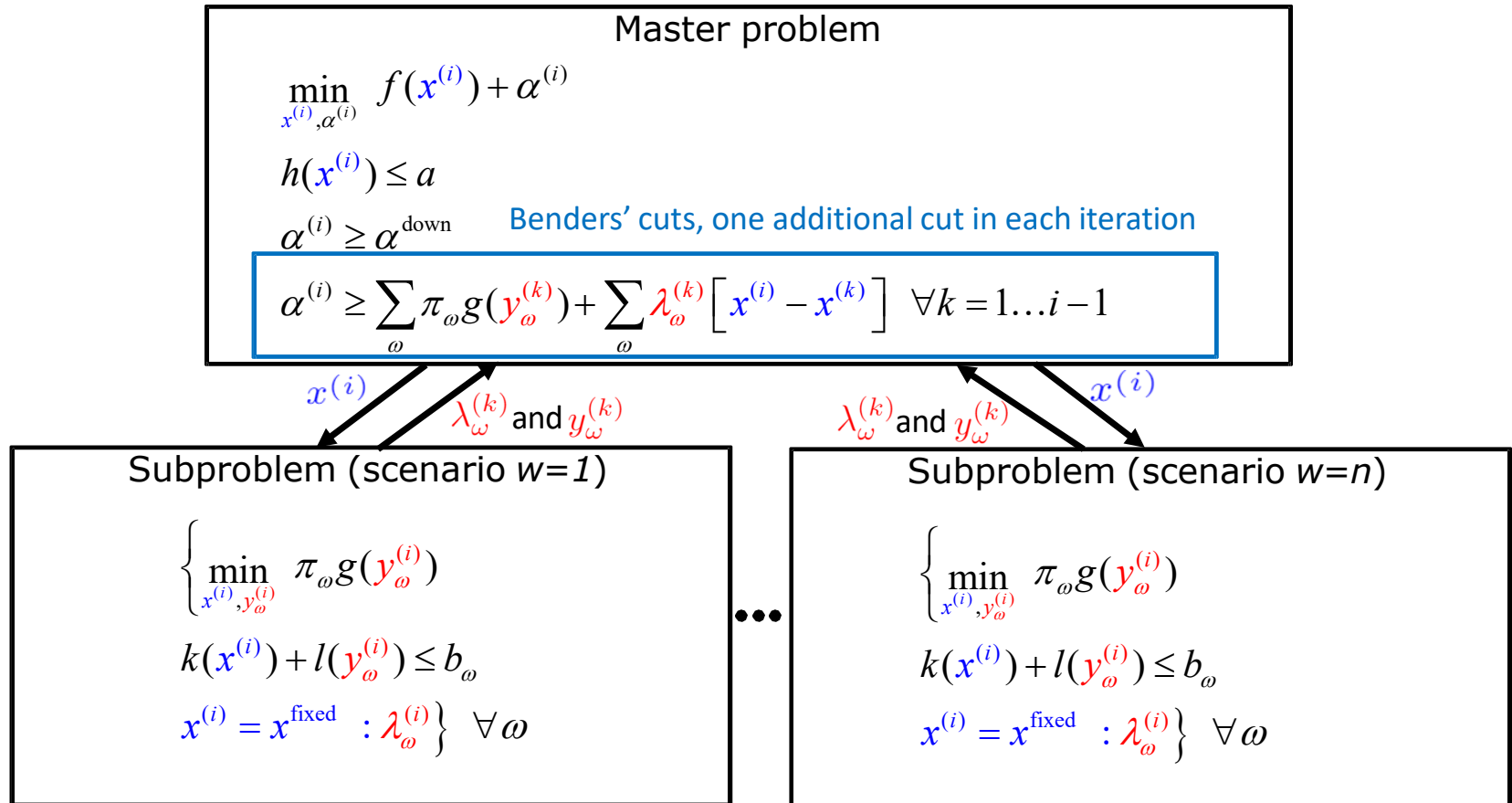
Two-stage stochastic program: Benders' solution



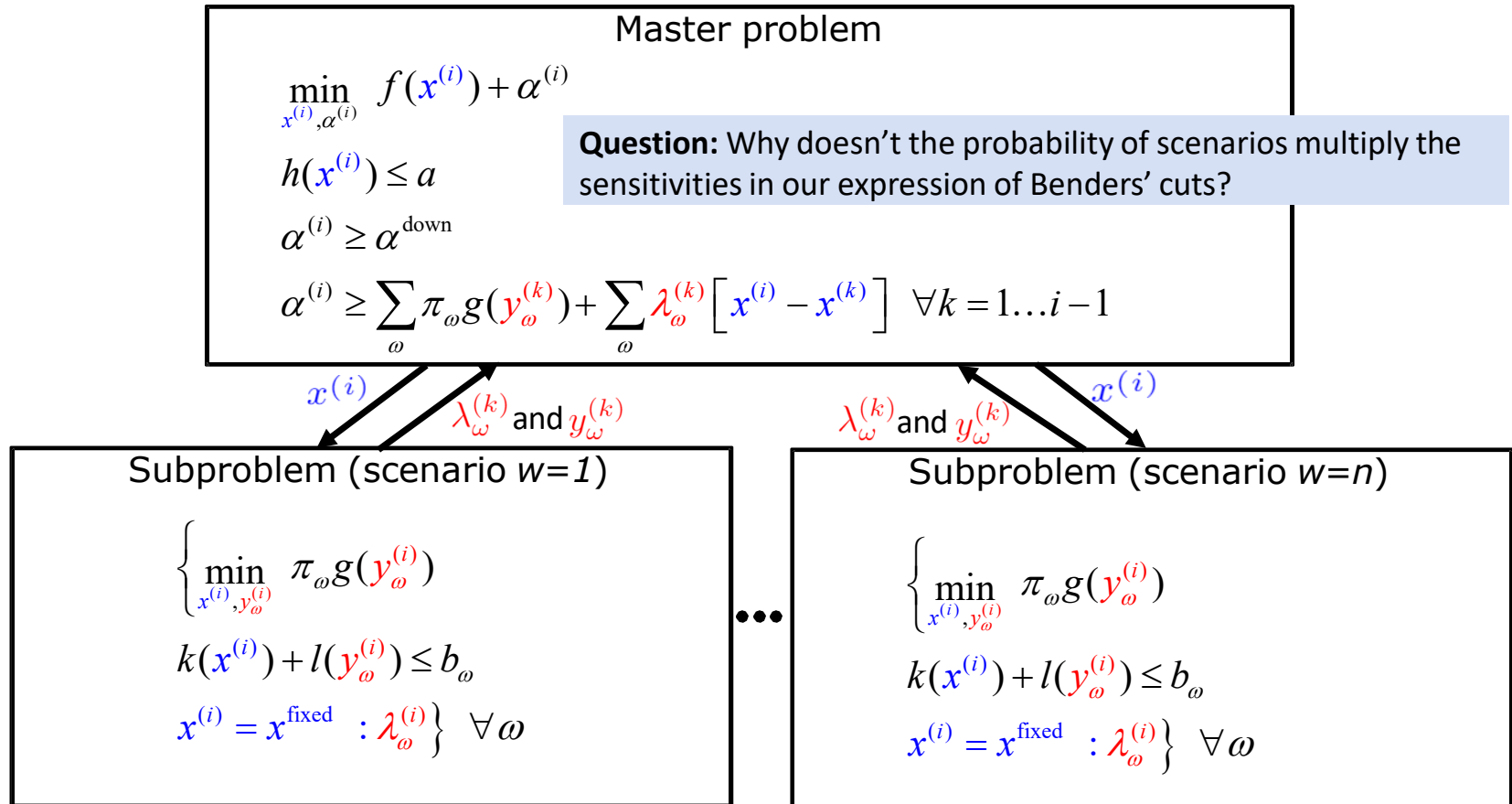
Two-stage stochastic program: Benders' solution



Two-stage stochastic program: Benders' solution



Two-stage stochastic program: Benders' solution



Decomposed model

Master problem

$$\begin{aligned}
 & \min_{x^{(i)}, \alpha^{(i)}} f(x^{(i)}) + \alpha^{(i)} \\
 & h(x^{(i)}) \leq a \\
 & \alpha^{(i)} \geq \alpha^{\text{down}} \\
 & \alpha^{(i)} \geq \sum_{\omega} \pi_{\omega} g(y_{\omega}^{(k)}) \\
 & \quad + \sum_{\omega} \lambda_{\omega}^{(k)} [x^{(i)} - x^{(k)}] \\
 & \quad \forall k = 1 \dots i-1
 \end{aligned}$$

Subproblem

$$\begin{aligned}
 & \left\{ \min_{x^{(i)}, y_{\omega}^{(i)}} \pi_{\omega} g(y_{\omega}^{(i)}) \right. \\
 & k(x^{(i)}) + l(y_{\omega}^{(i)}) \leq b_{\omega} \\
 & \left. x^{(i)} = x^{\text{fixed}} : \lambda_{\omega}^{(i)} \right\} \quad \forall \omega
 \end{aligned}$$

Decomposed model: Equivalent form

Master problem

$$\min_{x^{(i)}, \alpha^{(i)}} f(x^{(i)}) + \alpha^{(i)}$$

$$h(x^{(i)}) \leq a$$

$$\alpha^{(i)} \geq \alpha^{\text{down}}$$

$$\begin{aligned} \alpha^{(i)} &\geq \sum_{\omega} \pi_{\omega} g(y_{\omega}^{(k)}) \\ &\quad + \sum_{\omega} \pi_{\omega} \lambda_{\omega}^{(k)} [x^{(i)} - x^{(k)}] \\ &\quad \forall k = 1 \dots i-1 \end{aligned}$$

Subproblem

$$\left\{ \min_{x^{(i)}, y_{\omega}^{(i)}} g(y_{\omega}^{(i)}) \right.$$

$$k(x^{(i)}) + l(y_{\omega}^{(i)}) \leq b_{\omega}$$

$$x^{(i)} = x^{\text{fixed}} : \lambda_{\omega}^{(i)} \} \quad \forall \omega$$

Multi-cut Benders Decomposition

Concept

- Extension of classical Benders decomposition.
- Instead of one aggregated cut per iteration $\rightarrow$ one cut per subproblem.
- Useful in stochastic programming (one subproblem per scenario).

Master problem

$$\min_{x^{(i)}, \alpha^{(i)}} f(x^{(i)}) + \sum_{\omega} \pi_{\omega} \alpha_{\omega}$$

$$h(x^{(i)}) \leq a$$

$$\alpha_{\omega} \geq \alpha_{\omega}^{\text{down}} \quad \forall \omega \in \Omega$$

$$\alpha_{\omega} \geq g(y_{\omega}^{(k)})$$

$$+ \lambda_{\omega}^{(k)} [x^{(i)} - x^{(k)}]$$

$$\forall k = 1 \dots i-1, \forall \omega \in \Omega$$

Subproblem

$$\left\{ \min_{x^{(i)}, y_{\omega}^{(i)}} g(y_{\omega}^{(i)}) \right.$$

$$k(x^{(i)}) + l(y_{\omega}^{(i)}) \leq b_{\omega}$$

$$x^{(i)} = x^{\text{fixed}} : \lambda_{\omega}^{(i)} \left. \right\} \quad \forall \omega \in \Omega$$

Multi-cut Benders Decomposition

Advantages

- Provides richer information to the master problem.
- Faster convergence (fewer iterations).
- Parallelizable across scenarios.

Drawbacks

- Larger master problem (many cuts).
- Can be heavy with many scenarios.

Application Example

- Power systems: SCUC, stochastic OPF.
- Each contingency/scenario = one subproblem.
- Multi-cut Benders adds one cut per contingency.

Benders' decomposition – Stopping criterion

When can we consider that the algorithm has converged?

→ When the value of the **objective function of the “original problem”** in the subproblems and in the master problem has converged

We define at each iteration:

$$LB^{(i)} = f(x^{(i)}) + \alpha^{(i)}$$

$$UB^{(i)} = f(x^{(i)}) + \sum_{\omega} \pi_{\omega} g(y_{\omega}^{(i)}) \quad \leftarrow \text{The objective function of subproblem}$$

When $|LB^{(i)} - UB^{(i)}| \leq \varepsilon$, the algorithm has converged.

Stochastic market clearing: Benders' solution

- We assume that G_A can only deliver power to the day-ahead market
- The system operator commissioned wind power with the installed capacity of 80 MW

G_A	G_B	WP	Load
• Capacity: 100 MW • Production cost: 10 €/MWh • Inflexible generator	• Capacity: 40 MW • Production cost: 30 €/MWh • Flexible: reserve cap. Cost: 25 €/MW	• Capacity 80 MW • Production cost: 0 €/MWh	• 120 MW (inelastic and deterministic) • Rationing cost: 200 €/MWh

We assume that according to the best forecast for wind the realisation in a real-time production is 30 MW

Stochastic market clearing: Nomenclature

Indices and sets

$k \in 1, \dots, N_G$: generator index
 $s \in 1, \dots, N_S$: wind / real-time scenario
 N_G : number of generators
 N_S : number of scenarios

Parameters

C_k : generation marginal cost
 C_k^u, C_k^d : reserve capacity costs
 C^{rat} : load-rationing penalty
 π_s : scenario probability
 L : load demand
 $\bar{p}_k, \bar{w}$: generation and wind capacities
 w^{RT}_s : realised wind production

First-stage decisions

p_k : day-ahead production
 R_k^u, R_k^d : upward / downward reserve capacity
 w^{DA} : scheduled wind in day-ahead
 z : expected total system cost

Second-stage / scenario decisions

$r_{k,s}^u, r_{k,s}^d$: upward / downward reserve activation
 p_s^{rat} : rationed load
 w_s^{sp} : spilled wind power
 λ_s^{RT} : real-time balancing price

Stochastic market clearing: Benders' solution

$$\underset{p_k, R_k^u, R_k^d, w^{DA}, r_{k,s}^u, r_{k,s}^d, w_s^{sp}, p_s^{rat}}{\text{Minimize}} \quad z = \sum_{k=1}^{N_G} (c_k p_k + c_k^u R_k^u + c_k^d R_k^d) + \sum_{s=1}^{N_s} \pi_s \left(\sum_{k=1}^{N_G} c_k (r_{k,s}^u - r_{k,s}^d) + c^{rat} p_s^{rat} \right)$$

subject to

$$\sum_{k=1}^{N_G} p_k + w^{DA} = L, \quad (\lambda^{DA})$$

$$w^{DA} \leq \bar{w}$$

$$p_k + R_k^u \leq \bar{p}_k, \quad k = 1 \dots N_G$$

$$p_k - R_k^d \geq 0, \quad k = 1 \dots N_G$$

$$\sum_{k=1}^{N_G} (r_{k,s}^u - r_{k,s}^d) + p_s^{rat} + (w_s^{RT} - w^{DA} - w_s^{sp}) = 0, \quad s = 1 \dots N_s \quad (\lambda_s^{RT})$$

$$r_{k,s}^u \leq R_k^u, \quad k = 1 \dots N_G, \quad s = 1 \dots N_s$$

$$r_{k,s}^d \leq R_k^d, \quad k = 1 \dots N_G, \quad s = 1 \dots N_s$$

$$w_s^{sp} \leq w_s^{RT}, \quad s = 1 \dots N_s$$

$$p_s^{rat} \leq L, \quad s = 1 \dots N_s$$

Stochastic market clearing: Benders' solution

DA dispatch + allocated reserve capacity (first-stage decision variables)

$$\text{Minimize}_{p_k, R_k^u, R_k^d, w^{DA}, r_{k,s}^u, r_{k,s}^d, w_s^{sp}, p_s^{rat}} z = \sum_{k=1}^{N_G} (c_k p_k + c_k^u R_k^u + c_k^d R_k^d) + \sum_{s=1}^{N_s} \pi_s \left(\sum_{k=1}^{N_G} c_k (r_{k,s}^u - r_{k,s}^d) + c^{rat} p_s^{rat} \right)$$

subject to

$$\sum_{k=1}^{N_G} p_k + w^{DA} = l, \quad (\lambda^{DA})$$

$$w^{DA} \leq \bar{w}$$

$$p_k + R_k^u \leq \bar{p}_k, \quad k = 1 \dots N_G$$

$$p_k - R_k^d \geq 0, \quad k = 1 \dots N_G$$

$$\sum_{k=1}^{N_G} (r_{k,s}^u - r_{k,s}^d) + p_s^{rat} + (w_s^{RT} - w^{DA} - w_s^{sp}) = 0, \quad s = 1 \dots N_s \quad (\lambda_s^{RT})$$

$$r_{k,s}^u \leq R_k^u, \quad k = 1 \dots N_G, \quad s = 1 \dots N_s$$

$$r_{k,s}^d \leq R_k^d, \quad k = 1 \dots N_G, \quad s = 1 \dots N_s$$

$$w_s^{sp} \leq w_s^{RT}, \quad s = 1 \dots N_s$$

$$p_s^{rat} \leq L, \quad s = 1 \dots N_s$$

Mathematical formulation: Stochastic market-clearing

$$\underset{p_k, R_k^u, R_k^d, w^{DA}, r_{k,s}^u, r_{k,s}^d, w_s^{sp}, p_s^{rat}}{\text{Minimize}} \quad z = \underbrace{\sum_{k=1}^{N_G} (c_k p_k + c_k^u R_k^u + c_k^d R_k^d)}_{\text{Cost in the DA stage}} + \sum_{s=1}^{N_s} \pi_s \left(\sum_{k=1}^{N_G} c_k (r_{k,s}^u - r_{k,s}^d) + c^{rat} p_s^{rat} \right)$$

subject to

$$\begin{aligned} \sum_{k=1}^{N_G} p_k + w^{DA} &= L, \quad (\lambda^{DA}) \\ w^{DA} &\leq \bar{w} \\ p_k + R_k^u &\leq \bar{p}_k, \quad k = 1 \dots N_G \\ p_k - R_k^d &\geq 0, \quad k = 1 \dots N_G \end{aligned}$$

DA constraints

$$\begin{aligned} \sum_{k=1}^{N_G} (r_{k,s}^u - r_{k,s}^d) + p_s^{rat} + (w_s^{RT} - w^{DA} - w_s^{sp}) &= 0, \quad s = 1 \dots N_s \quad (\lambda_s^{RT}) \\ r_{k,s}^u &\leq R_k^u, \quad k = 1 \dots N_G, \quad s = 1 \dots N_s \\ r_{k,s}^d &\leq R_k^d, \quad k = 1 \dots N_G, \quad s = 1 \dots N_s \\ w_s^{sp} &\leq w_s^{RT}, \quad s = 1 \dots N_s \\ p_s^{rat} &\leq L, \quad s = 1 \dots N_s \end{aligned}$$

Mathematical formulation: Stochastic market-clearing

Real-time (RT) re-dispatch per scenario (second-stage decision variables)

$$\text{Minimize}_{p_k, R_k^u, R_k^d, w^{DA}, r_{k,s}^u, r_{k,s}^d, w_s^{sp}, p_s^{rat}} z = \sum_{k=1}^{N_G} (c_k p_k + c_k^u R_k^u + c_k^d R_k^d) + \sum_{s=1}^{N_s} \pi_s \left(\sum_{k=1}^{N_G} c_k (r_{k,s}^u - r_{k,s}^d) + c^{rat} p_s^{rat} \right)$$

subject to

$$\sum_{k=1}^{N_G} p_k + w^{DA} = L, \quad (\lambda^{DA})$$

$$w^{DA} \leq \bar{w}$$

$$p_k + R_k^u \leq \bar{p}_k, \quad k = 1 \dots N_G$$

$$p_k - R_k^d \geq 0, \quad k = 1 \dots N_G$$

$$\sum_{k=1}^{N_G} (r_{k,s}^u - r_{k,s}^d) + p_s^{rat} + (w_s^{RT} - w^{DA} - w_s^{sp}) = 0, \quad s = 1 \dots N_s \quad (\lambda_s^{RT})$$

$$r_{k,s}^u \leq R_k^u, \quad k = 1 \dots N_G, \quad s = 1 \dots N_s$$

$$r_{k,s}^d \leq R_k^d, \quad k = 1 \dots N_G, \quad s = 1 \dots N_s$$

$$w_s^{sp} \leq w_s^{RT}, \quad s = 1 \dots N_s$$

$$p_s^{rat} \leq L, \quad s = 1 \dots N_s$$

Stochastic market clearing: Benders' solution

$$\text{Minimize}_{p_k, R_k^u, R_k^d, w^{DA}, r_{k,s}^u, r_{k,s}^d, w_s^{sp}, p_s^{rat}} z = \sum_{k=1}^{N_G} (c_k p_k + c_k^u R_k^u + c_k^d R_k^d) + \sum_{s=1}^{N_s} \pi_s \left(\sum_{k=1}^{N_G} c_k (r_{k,s}^u - r_{k,s}^d) + c^{rat} p_s^{rat} \right)$$

Expected re-dispatch cost in the RT stage

subject to

$$\sum_{k=1}^{N_G} p_k + w^{DA} = L, \quad (\lambda^{DA})$$

$$w^{DA} \leq \bar{w}$$

$$p_k + R_k^u \leq \bar{p}_k, \quad k = 1 \dots N_G$$

$$p_k - R_k^d \geq 0, \quad k = 1 \dots N_G$$

$$\sum_{k=1}^{N_G} (r_{k,s}^u - r_{k,s}^d) + p_s^{rat} + (w_s^{RT} - w^{DA} - w_s^{sp}) = 0, \quad s = 1 \dots N_s \quad (\lambda_s^{RT})$$

$$r_{k,s}^u \leq R_k^u, \quad k = 1 \dots N_G, \quad s = 1 \dots N_s$$

$$r_{k,s}^d \leq R_k^d, \quad k = 1 \dots N_G, \quad s = 1 \dots N_s$$

$$w_s^{sp} \leq w_s^{RT}, \quad s = 1 \dots N_s$$

$$p_s^{rat} \leq L, \quad s = 1 \dots N_s$$

RT constraints under scenarios s1 to s4

Stochastic market clearing: Benders' solution

Complicating variables: DA dispatch of generators and wind farm + allocated reserve capacity

$$\text{Minimize}_{p_k, R_k^u, R_k^d, w^{DA}, r_{k,s}^u, r_{k,s}^d, w_s^{sp}, p_s^{rat}} z = \sum_{k=1}^{N_G} (c_k p_k + c_k^u R_k^u + c_k^d R_k^d) + \sum_{s=1}^{N_s} \pi_s \left(\sum_{k=1}^{N_G} c_k (r_{k,s}^u - r_{k,s}^d) + c^{rat} p_s^{rat} \right)$$

subject to

$$\sum_{k=1}^{N_G} p_k + w^{DA} = L, \quad (\lambda^{DA})$$

$$w^{DA} \leq \bar{w}$$

$$p_k + R_k^u \leq \bar{p}_k, \quad k = 1 \dots N_G$$

$$p_k - R_k^d \geq 0, \quad k = 1 \dots N_G$$

$$\sum_{k=1}^{N_G} (r_{k,s}^u - r_{k,s}^d) + p_s^{rat} + (w_s^{RT} - w^{DA} - w_s^{sp}) = 0, \quad s = 1 \dots N_s \quad (\lambda_s^{RT})$$

$$r_{k,s}^u \leq R_k^u, \quad k = 1 \dots N_G, \quad s = 1 \dots N_s$$

$$r_{k,s}^d \leq R_k^d, \quad k = 1 \dots N_G, \quad s = 1 \dots N_s$$

$$w_s^{sp} \leq w_s^{RT}, \quad s = 1 \dots N_s$$

$$p_s^{rat} \leq L, \quad s = 1 \dots N_s$$

Stochastic market clearing: Benders' solution

Subproblem:

$$\text{Minimize}_{p_k^{(i)}, R_k^{u(i)}, R_k^{d(i)}, w^{DA(i)}, r_{k,s}^u, r_{k,s}^d, w_s^{sp}, p_s^{rat}} \sum_{s=1}^{N_s} \pi_s \left(\sum_{k=1}^{N_G} c_k (r_{k,s}^u - r_{k,s}^d) + c^{rat} p_s^{rat} \right)$$

subject to

$$\sum_{k=1}^{N_G} (r_{k,s}^u - r_{k,s}^d) + p_s^{rat} + (w_s^{RT} - w^{DA(i)} - w_s^{sp}) = 0, \quad s = 1 \dots N_s \quad (\lambda_s^{RT})$$

$$r_{k,s}^u \leq R_k^{u(i)}, \quad k = 1 \dots N_G, \quad s = 1 \dots N_s$$

$$r_{k,s}^d \leq R_k^{d(i)}, \quad k = 1 \dots N_G, \quad s = 1 \dots N_s$$

$$w_s^{SP} \leq w_s^{RT}, \quad s = 1 \dots N_s$$

$$p_s^{rat} \leq L, \quad s = 1 \dots N_s$$

$$R_k^{u(i)} = R_k^{u, \text{fixed}(i)}, \quad k = 1 \dots N_G, \quad s = 1 \dots N_s$$

$$R_k^{d(i)} = R_k^{d, \text{fixed}(i)}, \quad k = 1 \dots N_G, \quad s = 1 \dots N_s$$

$$w^{DA(i)} = w^{DA, \text{fixed}(i)}, \quad k = 1 \dots N_G, \quad s = 1 \dots N_s$$

$$p_k^{(i)} = p_k^{\text{fixed}(i)}, \quad k = 1 \dots N_G, \quad s = 1 \dots N_s$$

$$(\rho_{k,s}^{Ru(i)})$$

$$(\rho_{k,s}^{Rd(i)})$$

$$(\rho_{k,s}^{w(i)})$$

$$(\rho_{k,s}^{p(i)})$$

Sensitivities of subproblem under scenarios s1 to s4, with respect to all complicating variables

Stochastic market clearing: Benders' solution

Master problem:

$$\text{Minimize}_{p_k^{(i)}, R_k^{u(i)}, R_k^{d(i)}, w^{DA(i)}, \alpha^{(i)}} \sum_{k=1}^{N_G} \left(c_k p_k^{(i)} + c_k^u R_k^{u(i)} + c_k^d R_k^{d(i)} \right) + \alpha^{(i)}$$

subject to

$$\sum_{k=1}^{N_G} p_k^{(i)} + w^{DA(i)} = L, \quad (\lambda^{DA})$$

$$w^{DA(i)} \leq \bar{w}$$

$$p_k^{(i)} + R_k^{u(i)} \leq \bar{p}_k, \quad k = 1 \dots N_G$$

$$p_k^{(i)} - R_k^{d(i)} \geq 0, \quad k = 1 \dots N_G$$

$$\alpha^{(i)} \geq \alpha^{down}$$

Benders' cuts?

Stochastic market clearing: Benders' solution

Master problem:

$$\text{Minimize}_{p_k^{(i)}, R_k^{u(i)}, R_k^{d(i)}, w^{DA(i)}, \alpha^{(i)}} \sum_{k=1}^{N_G} \left(c_k p_k^{(i)} + c_k^u R_k^{u(i)} + c_k^d R_k^{d(i)} \right) + \alpha^{(i)}$$

subject to

$$\sum_{k=1}^{N_G} p_k^{(i)} + w^{DA(i)} = L, \quad (\lambda^{DA})$$

$$w^{DA(i)} \leq \bar{w}$$

$$p_k^{(i)} + R_k^{u(i)} \leq \bar{p}_k, \quad k = 1 \dots N_G$$

$$p_k^{(i)} - R_k^{d(i)} \geq 0, \quad k = 1 \dots N_G$$

$$\alpha^{(i)} \geq \alpha^{down}$$

$$\begin{aligned} \alpha^{(i)} \geq & \sum_{s=1}^{N_s} \pi_s \left(\sum_{k=1}^{N_G} c_k (r_{k,s}^u - r_{k,s}^d) + c^{rat} p_s^{rat} \right) \quad \leftarrow \text{Value from the subproblem of the previous} \\ & \text{iteration } j \text{ (} r, p^{rat} \text{ are numbers here, not variables)} \\ & + \sum_{k=1}^{N_G} \sum_{s=1}^{N_s} \rho_{k,s}^{Ru(j)} \left(R_k^{u(i)} - R_k^{u(j)} \right) + \sum_{k=1}^{N_G} \sum_{s=1}^{N_s} \rho_{k,s}^{Rd(j)} \left(R_k^{d(i)} - R_k^{d(j)} \right) \\ & + \sum_{s=1}^{N_s} \rho_s^{w(j)} \left(w^{DA(i)} - w^{DA(j)} \right) + \sum_{k=1}^{N_G} \sum_{s=1}^{N_s} \rho_{k,s}^{p(j)} \left(p_k^{(i)} - p_k^{(j)} \right), \quad j = 1, \dots, i-1 \end{aligned}$$

Stochastic market clearing: Benders' solution

Master problem:

$$\text{Minimize}_{p_k^{(i)}, R_k^{u(i)}, R_k^{d(i)}, w^{DA(i)}, \alpha^{(i)}} \sum_{k=1}^{N_G} \left(c_k p_k^{(i)} + c_k^u R_k^{u(i)} + c_k^d R_k^{d(i)} \right) + \alpha^{(i)}$$

subject to

$$\sum_{k=1}^{N_G} p_k^{(i)} + w^{DA(i)} = L, \quad (\lambda^{DA})$$

$$w^{DA(i)} \leq \bar{w}$$

$$p_k^{(i)} + R_k^{u(i)} \leq \bar{p}_k, \quad k = 1 \dots N_G$$

$$p_k^{(i)} - R_k^{d(i)} \geq 0, \quad k = 1 \dots N_G$$

$$\alpha^{(i)} \geq \alpha^{down}$$

$$\alpha^{(i)} \geq \sum_{s=1}^{N_s} \pi_s \left(\sum_{k=1}^{N_G} c_k \left(r_{k,s}^u - r_{k,s}^d \right) + c^{rat} p_s^{rat} \right)$$

Sensitivities of subproblem under scenarios s1 to s4,
with respect to complicating variables R_k^u

$$- \sum_{k=1}^{N_G} \sum_{s=1}^{N_s} \rho_{k,s}^{Ru(j)} \left(R_k^{u(i)} - R_k^{u(j)} \right) - \sum_{k=1}^{N_G} \sum_{s=1}^{N_s} \rho_{k,s}^{Rd(j)} \left(R_k^{d(i)} - R_k^{d(j)} \right)$$

$$+ \sum_{s=1}^{N_s} \rho_s^{w(j)} \left(w^{DA(i)} - w^{DA(j)} \right) + \sum_{k=1}^{N_G} \sum_{s=1}^{N_s} \rho_{k,s}^{p(j)} \left(p_k^{(i)} - p_k^{(j)} \right), \quad j = 1, \dots, i-1$$

Stochastic market clearing: Benders' solution

Master problem:

$$\text{Minimize}_{p_k^{(i)}, R_k^{u(i)}, R_k^{d(i)}, w^{DA(i)}, \alpha^{(i)}} \sum_{k=1}^{N_G} \left(c_k p_k^{(i)} + c_k^u R_k^{u(i)} + c_k^d R_k^{d(i)} \right) + \alpha^{(i)}$$

subject to

$$\sum_{k=1}^{N_G} p_k^{(i)} + w^{DA(i)} = L, \quad (\lambda^{DA})$$

$$w^{DA(i)} \leq \bar{w}$$

$$p_k^{(i)} + R_k^{u(i)} \leq \bar{p}_k, \quad k = 1 \dots N_G$$

$$p_k^{(i)} - R_k^{d(i)} \geq 0, \quad k = 1 \dots N_G$$

$$\alpha^{(i)} \geq \alpha^{down}$$

$$\alpha^{(i)} \geq \sum_{s=1}^{N_s} \pi_s \left(\sum_{k=1}^{N_G} c_k \left(r_{k,s}^u - r_{k,s}^d \right) + c^{rat} p_s^{rat} \right)$$

$$+ \sum_{k=1}^{N_G} \sum_{s=1}^{N_s} \rho_{k,s}^{Ru(j)} \left(R_k^{u(i)} - R_k^{u(j)} \right) - \sum_{k=1}^{N_G} \sum_{s=1}^{N_s} \rho_{k,s}^{Rd(j)} \left(R_k^{d(i)} - R_k^{d(j)} \right)$$

$$+ \sum_{s=1}^{N_s} \rho_s^{w(j)} \left(w^{DA(i)} - w^{DA(j)} \right) + \sum_{k=1}^{N_G} \sum_{s=1}^{N_s} \rho_{k,s}^{p(j)} \left(p_k^{(i)} - p_k^{(j)} \right), \quad j = 1, \dots, i-1$$

Sensitivities of subproblem under scenarios s1 to s4,
with respect to complicating variables R_k^d

Stochastic market clearing: Benders' solution

Master problem:

$$\text{Minimize}_{p_k^{(i)}, R_k^{u(i)}, R_k^{d(i)}, w^{DA(i)}, \alpha^{(i)}} \sum_{k=1}^{N_G} \left(c_k p_k^{(i)} + c_k^u R_k^{u(i)} + c_k^d R_k^{d(i)} \right) + \alpha^{(i)}$$

subject to

$$\sum_{k=1}^{N_G} p_k^{(i)} + w^{DA(i)} = L, \quad (\lambda^{DA})$$

$$w^{DA(i)} \leq \bar{w}$$

$$p_k^{(i)} + R_k^{u(i)} \leq \bar{p}_k, \quad k = 1 \dots N_G$$

$$p_k^{(i)} - R_k^{d(i)} \geq 0, \quad k = 1 \dots N_G$$

$$\alpha^{(i)} \geq \alpha^{down}$$

$$\alpha^{(i)} \geq \sum_{s=1}^{N_s} \pi_s \left(\sum_{k=1}^{N_G} c_k \left(r_{k,s}^u - r_{k,s}^d \right) + c^{rat} p_s^{rat} \right)$$

$$+ \sum_{k=1}^{N_G} \sum_{s=1}^{N_s} \rho_{k,s}^{Ru(j)} \left(R_k^{u(i)} - R_k^{u(j)} \right) + \sum_{k=1}^{N_G} \sum_{s=1}^{N_s} \rho_{k,s}^{Rd(j)} \left(R_k^{d(i)} - R_k^{d(j)} \right)$$

$$+ \sum_{s=1}^{N_s} \rho_s^{w(j)} \left(w^{DA(i)} - w^{DA(j)} \right) + \sum_{k=1}^{N_G} \sum_{s=1}^{N_s} \rho_{k,s}^{p(j)} \left(p_k^{(i)} - p_k^{(j)} \right), \quad j = 1, \dots, i-1$$

Sensitivities of subproblem under scenarios s1 to s4, with respect to complicating variable w^{DA}

Stochastic market clearing: Benders' solution

Master problem:

$$\text{Minimize}_{p_k^{(i)}, R_k^{u(i)}, R_k^{d(i)}, w^{DA(i)}, \alpha^{(i)}} \sum_{k=1}^{N_G} \left(c_k p_k^{(i)} + c_k^u R_k^{u(i)} + c_k^d R_k^{d(i)} \right) + \alpha^{(i)}$$

subject to

$$\sum_{k=1}^{N_G} p_k^{(i)} + w^{DA(i)} = l, \quad (\lambda^{DA})$$

$$w^{DA(i)} \leq \bar{w}$$

$$p_k^{(i)} + R_k^{u(i)} \leq \bar{p}_k, \quad k = 1 \dots N_G$$

$$p_k^{(i)} - R_k^{d(i)} \geq 0, \quad k = 1 \dots N_G$$

$$\alpha^{(i)} \geq \alpha^{down}$$

$$\begin{aligned} \alpha^{(i)} \geq & \sum_{s=1}^{N_s} \pi_s \left(\sum_{k=1}^{N_G} c_k (r_{k,s}^u - r_{k,s}^d) + c^{rat} p_s^{rat} \right) \\ & + \sum_{k=1}^{N_G} \sum_{s=1}^{N_s} \rho_{k,s}^{Ru(j)} (R_k^{u(i)} - R_k^{u(j)}) + \sum_{k=1}^{N_G} \sum_{s=1}^{N_s} \rho_{k,s}^{Rd(j)} (R_k^{d(i)} - R_k^{d(j)}) \\ & + \sum_{s=1}^{N_s} \rho_s^{w(j)} (w^{DA(i)} - w^{DA(j)}) + \sum_{k=1}^{N_G} \sum_{s=1}^{N_s} \rho_{k,s}^{p(j)} (p_k^{(i)} - p_k^{(j)}) \quad j = 1, \dots, i-1 \end{aligned}$$

Sensitivities of subproblem under scenarios s1 to s4, with respect to complicating variable p_k

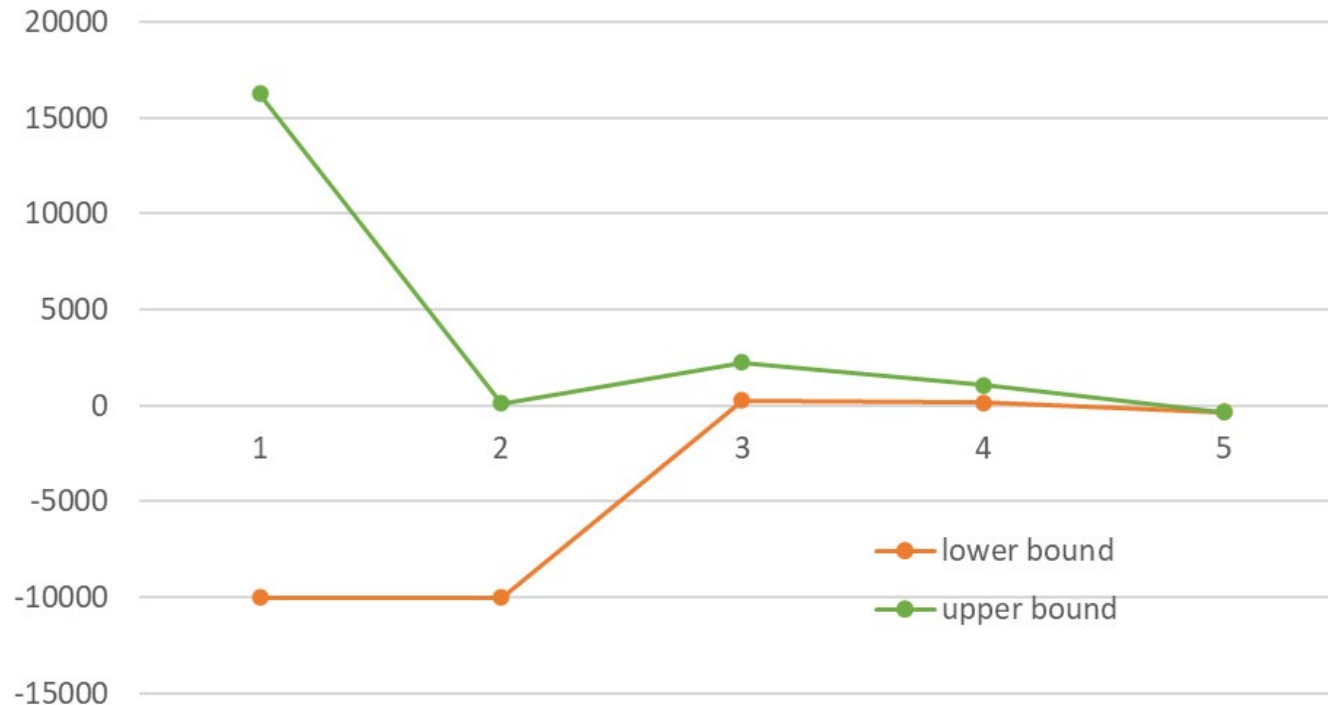
Stochastic market-clearing

	DA+R	RT (scenario 1) Wind power realization: 30 MW	RT (scenario 2) Wind power realization: 40 MW	RT (scenario 3) Wind power realization: 80 MW	RT (scenario 4) Wind power realization: 5 MW
G_A [MW]	100	unchanged	unchanged	unchanged	unchanged
G_B [MW]	15	-15 (down regulation)	-15 (down regulation)	-15 (down regulation)	unchanged
WP [MW]	5	+15 (10 MW spilled)	+15 (20 MW spilled)	+15 (60 MW spilled)	unchanged
R_B^u [MW]	0	-	-	-	-
R_B^d [MW]	15	-15	-15	-15	-
Load [MW]	120	unchanged (not curtailed)	unchanged (not curtailed)	unchanged (not curtailed)	unchanged (not curtailed)
Corresponding operational cost [€]	1600	-450	-450	-450	0

Total expected operational cost of the system

$$= 1600 + 0.25 * [-450 - 450 - 450 + 0] = 1262.5 \text{ €}$$

Benders algorithm Convergence



$$UB^{(i)} = \underset{p_k^{(i)}, R_k^{u(i)}, R_k^{d(i)}, w^{DA(i)}, r_{k,s}^u, r_{k,s}^d, w_s^{sp}, p_s^{rat}}{\text{Minimize}} \sum_{s=1}^{N_s} \pi_s \left(\sum_{k=1}^{N_G} c_k (r_{k,s}^u - r_{k,s}^d) + c^{rat} p_s^{rat} \right)$$

$$LB^{(i)} = \alpha^{(i)}$$

Both bounds are shown without the first-stage cost, which is common to LB and UB, so the gap is the same. They meet at iteration 5 (-337.5 €).

...To sum up

- Stochastic optimization problems are decomposable
- Complicating variables: first-stage (here-and-now) decision variables
- Number of subproblems: one per scenario (at least)
- We have expressed the Benders' algorithm for a generic (linear) stochastic optimization problem
- We applied Benders' decomposition to stochastic market - clearing